Lithos 286–287 (2017) 264–301

Contents lists available at ScienceDirect

Lithos

journal homepage: www.elsevier.com/locate/lithos

Invited synthesis paper

The Cenozoic magmatism of East-Africa: Part I — Flood basalts and pulsed magmatism

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ARTICLE INFO

Article history:
Received 12 March 2017
Accepted 22 May 2017
Available online 3 June 2017

Keywords:
East Africa
Magmatism
Plumes
LLSVP

ABSTRACT

Cenozoic magmatism in East Africa results from the interplay between lithospheric extension and material upwelling from the African Large Low Shear Velocity Province (LLSVP). The modern focusing of East African magmatism into oceanic spreading centers and continental rifts highlights the modern control of lithospheric thinning in magma generation processes, however the widespread, and volumetrically significant flood basalt events of the Eocene to Early Miocene suggest a significant role for material upwelling from the African LLSVP. The slow relative motion of the African plate during the Cenozoic has resulted in significant spatial overlap in lavas derived from different magmatic events. This complexity is being resolved with enhanced geochronological precision and a focus on the geochemical characteristics of the volcanic products. It is now apparent that there are three distinct pulses of basaltic volcanism, followed by either bimodal lavas or silicic volcanic products during this period: (A) Eocene Initial Phase from 45 to 34 Ma. This is a period of dominantly basaltic volcanism focused in Southern Ethiopia and Northern Kenya (Turkana). (B) Oligocene Traps phase from ~33.9 to 27 Ma. This period coincides with a significant increase in the aerial extent of volcanism with broadly age equivalent 1 to 2 km thick sequences of dominantly basalt centered on the NW Ethiopian Plateau and Yemen, (C) Early Miocene resurgence phase from ~26.9 to 22 Ma. This resurgence in basaltic volcanism is seen throughout the region at ca. 24–23 Ma, but is less volumetrically significant than the prior two basaltic pulses. With our developing understanding of the persistence of LLSVP anomalies within the mantle, I propose that the three basaltic pulses are ostensibly manifestations of the same plume–lithosphere interaction, requiring revision to the duration, magmatic extent, and magma volume of the African–Arabian Large Igneous Province.

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1. Introduction

1.1. The African LLSVP in global geodynamic context

The deep structure of the Earth's mantle is dominated by two geophysically defined zones where seismic waves travel slowly (Fig. 1), termed Large Low Shear Velocity Provinces (LLSVP: Garnero et al., 2016). These LLSVPs are regions of dense material that are located in the deep mantle beneath the Pacific and African plates (Garnero and McNamara, 2008). The magnitude of these structures hints at their persistence, potentially playing a role in the generation of large igneous provinces throughout the Phanerozoic (Torsvik et al., 2008). While geophysical techniques provide important constraints as to the geometry and density of these structures (Fig. 1), their origin may be probed by examining the magmas derived from the melting of material upwelling from these anomalies (Weis et al., 2011). Such studies are revealing tentative linkages between the existence of the LLSVPs and processes of early Earth accretion (Rizo et al., 2016).

Melt derived from materials rising from the Pacific LLSVP interact with the overlying Pacific plate, resulting in the creation of hotspot tracks throughout the region (Konter et al., 2008). While 'hotspot highways' exist where multiple tracks may overlap (Jackson et al., 2010), the typical spatial and temporal separation of these lavas has allowed for insights into the zoning of plume tracks (Frey et al., 2016), which have been attributed to heterogeneities within the LLSVP from which they are derived (Huang et al., 2011). Similar zoning may be observed in oceanic islands surrounding the African continent (Schwindrofska et al., 2016), however the relatively stationary African plate and the existence of thick continental lithosphere has complicated interpretations of magmas derived from the African LLSVP. Specifically, the temporal and spatial overlap of magmatic activity on the African plate requires detailed stratigraphic constraints (Fig. 2). This contribution is the first in a series examining magmatism in East Africa during the late Cenozoic (45 Ma to present) — one of the most pronounced manifestations of melt derived from material upwelling from the African LLSVP.

1.2. Background

Since exploration by European naturalists in the late 18th and throughout the 19th centuries, recorded accounts of the distribution

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http://dx.doi.org/10.1016/j.lithos.2017.05.014
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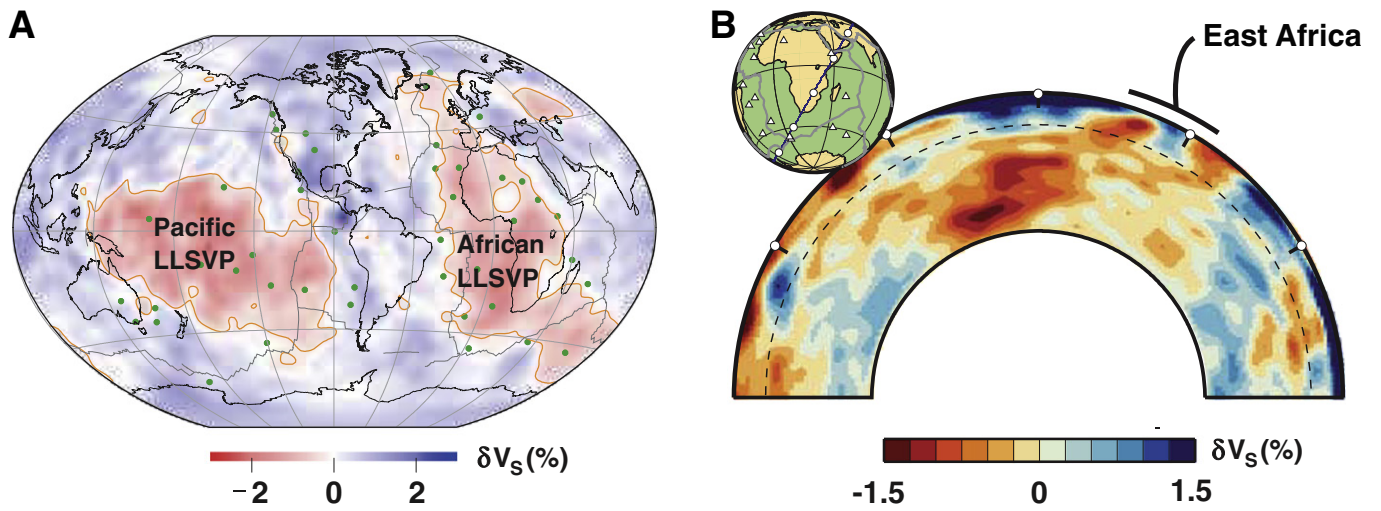


Fig. 1. A. Mantle shear velocity perturbation (δV_S) for the deep mantle (2800 km) derived from the seismic shear velocity model for the mantle (Garnero et al., 2016; Ritsema et al., 2011). The orange line represents the 30% of the core–mantle boundary area (Garnero et al., 2016). The African and Pacific Large Low Shear Velocity Provinces (LLSVP) are labeled. Hotspots (shown as green dots). B. Whole mantle cross section using the S40RTS seismic shear velocity model for the mantle showing vertical structure of the African LLSVP (Ritsema et al., 2011).

of magmatic rocks in the region began to accumulate within the scientific literature. Among the most celebrated of these accounts is that of John Gregory (Gregory, 1896), but this work was preceded and succeeded by a vast number of other works that is summarized by Mohr (2009). The motivation for this review is to place the body of existing knowledge on the distribution of magmatism in East Africa within a modern temporal context. Much of the published stratigraphic work creates either an array of localized basin-scale stratigraphic sections (e.g., McDougall and Brown, 2009), or attempts to correlate locally-derived volcanic stratigraphy more broadly (e.g., Zanettin et al., 1980). Both methods of volcanic unit mapping are predicated on a top-down approach to magmatic processes whereby volcanic products are treated in a similar fashion to sedimentary deposits in that they are used to solve temporal issues in the stratigraphic record (e.g., McDougall et al., 2012), or other near-surface or basin-scale questions. With the developments in high-precision dating and igneous geochemistry, a revised approach to the volcanic stratigraphy of East Africa is required. Within this review, I assemble the rich existing stratigraphic framework into broad divisions that are useful in examining magmatic processes that are occurring beneath the African Plate. This compilation is a crucial first step in unraveling the complex magmatic history of the region in order to probe the role of the African LLSVP in the generation of magmatism in East Africa over the past 45 Ma.

This review has been divided into time periods that are equivalent to the most significant magmatic events in East Africa during the Cenozoic. These periods are not necessarily separated by any eruptive hiatus, but they signify major changes in the style or extent of magmatism. This first contribution examines the time interval from the initiation of Cenozoic magmatism at ca. 45 Ma, and continues through to the termination of what I consider is plume-dominated magmatic activity ca. 22 Ma. Over the temporal interval examined within this contribution, three distinctive periods (Fig. 3) have been identified:

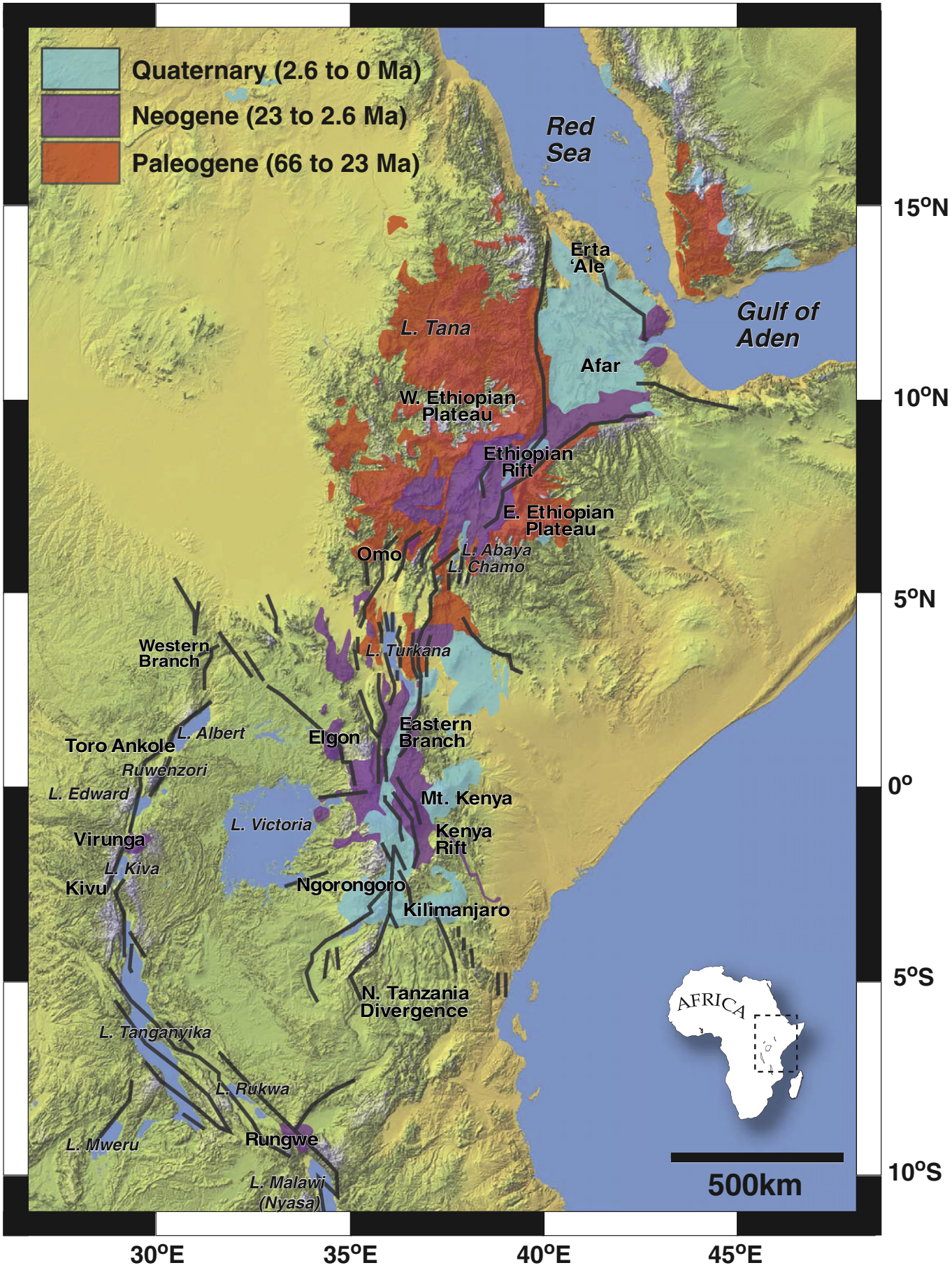
A) Eocene Initial Phase — from 45 to 34 Ma.
   This is a period of dominantly basaltic volcanism focused in Southern Ethiopia and Northern Kenya (Turkana). On the basis of our current geochronological understanding, these basalts are limited in aerial extent to those regions mentioned above and are therefore distinct from the overlying and much more spatially extensive early Oligocene rocks (e.g., Furman et al., 2006b; George and Rogers, 2002)
B) Oligocene Traps phase — from ~33.9 to 27 Ma
   This period coincides with a significant increase in the aerial extent of volcanism with broadly age equivalent 1 to 2 km thick sequences of dominantly basalt that is observed from Turkana in the south, to Yemen in the north, to Sudan in the west, and the margin of the Somali Plateau to the east. This is the period commonly attributed to the Ethiopian–Arabian Plateau flood basalts (Baker et al., 1996a; Hofmann et al., 1997; Rochette et al., 1998). Flood basalt activity is subsequently followed by a period of relative quiescence with magmatic activity focused dominantly within Turkana (Brown and McDougall, 2011), with localized silicic eruptions (Ukstins-Peate et al., 2008) and diking along the evolving rift margin in Ethiopia and Yemen (Rooney et al., 2013).
C) Early Miocene resurgence phase — from ~26.9 to 22 Ma
   The final period examined in this review is resurgence in basaltic volcanism that is seen throughout the region, but that is somewhat synchronous at ca. 24–23 Ma. This activity manifests as: Fissural basalts on the Somali Plateau; shield building events on the NW Ethiopian Plateau, and increased basaltic activity in Turkana (Brown and McDougall, 2011; Furman et al., 2006b; Kieffer et al., 2004). Magmatic activity during this period appears to be largely absent in Yemen but may be present along the Red Sea margin.

These periods are broadly defined and relate to specific magmatic events in the region. Given the difficulties of K–Ar dating in rocks from this region and the long time spans of eruption, I assign rocks to these divisions on the basis of age and stratigraphic relationships to the events.

2. The Eocene Initial Phase

2.1. Nomenclature and age constraints

Cenozoic magmatic activity in East Africa commenced ca. 45 Ma in Southern Ethiopia with the eruption of the Amaro basalts over a suspected Mesozoic non-fossiliferous fluviatile 'grit', which itself lies upon the Pan-African crystalline basement (Ebinger et al., 1993). The geographic extent of this early magmatic activity was somewhat unclear due to the difficulties of dating these rocks with traditional K–Ar methods. Accordingly, a pot-pourri of unit formations developed with unclear associations and ages, though the oldest units were considered to be limited to the SW Ethiopian


Quaternary (2.6 to 0 Ma)
Neogene (23 to 2.6 Ma)
Paleogene (66 to 23 Ma)
Red Sea
Gulf of Aden
Erta 'Ale
L. Tana
Afar
W. Ethiopian Plateau
Ethiopian Rift
E. Ethiopian Plateau
Omo
L. Abaya
L. Chamo
L. Turkana
Western Branch
Eastern Branch
Elgon
Toro Ankole
L. Albert
Ruwenzori
L. Edward
Virunga
L. Kiva
Kivu
L. Victoria
Mt. Kenya
Kenya Rift
Ngorongoro
Kilimanjaro
N. Tanzania Divergence
L. Tanganyika
L. Rukwa
L. Mweru
Rungwe
L. Malawi (Nyasa)
AFRICA
500km
15°N
10°N
5°N
0°
5°S
10°S
30°E
35°E
40°E
45°E

Plateau (Berhe et al., 1987; Davidson and Rex, 1980; Zanettin, 1978). With the application of ^{40}Ar–^{39}Ar methods on rocks from around the Chew Bahir basin and Amaro Horst (Fig. 4) it became apparent that the earliest stages of magmatism were much more broadly distributed and extended throughout much of Southern Ethiopia (Ebinger et al., 1993; Woldegabriel et al., 1991). These studies confirmed that the initial eruption of the Amaro basalts was geographically widespread, and occurred over an extended period lasting ~5 Ma (45.2–39.8 Ma; George et al., 1998) — see Section 7 for stratigraphic constraints. Towards the top of the Amaro basalts are the localized Arba Minch tuffs (exposed along the Chamo border fault). Conformably overlying these units are the Gamo Basalts (39.8 Ma–34.1 Ma), within which occurs an important marker bed — the widespread Amaro Tuff (36.9 Ma) (Ebinger et al., 1993; George et al., 1998).

There exists a significant diversity in unit names for Eocene rocks from SW Ethiopia: Geba basalts (Berhe et al., 1987), Omo basalts (Abbate and Sagri, 1980), Main Series basalts (Berhe et al., 1987; Davidson, 1983), and the Jimma Volcanics (Tefera et al., 1996). I propose the use of the 'Amaro/Gamo unit' for all Eocene volcanic rocks erupted throughout SW Ethiopia, given the robust temporal constraints available for the Amaro/Gamo lavas. The Eocene basalts of Southern Ethiopia shown in Fig. 4 are that of the 'Jimma Volcanics' as mapped by Tefera et al. (1996), which are interpreted as Eocene to Late Oligocene. These volcanic rocks are equivalent (where overlap occurs) to the Eocene–Oligocene 'Main Series' from the Omo River (Davidson, 1983). On the basis of the temporal constraints from the Amaro/Gamo lavas, I assign the entire unit to the Eocene but note that more $^{40}Ar/^{39}Ar$ dating is needed in the region from 7 to 8°N.

The Amaro/Gamo unit as now defined above is comprised of a package of largely transitional to tholeiitic basalts and associated silicic rocks (e.g. Davidson, 1983). An exception to this is a small exposure of alkaline rocks termed the 'Akobo basalts' which were noted by Davidson (1983) to be different to the overlying main package in being clinopyroxene phyric and consistently older than the voluminous transitional-tholeiitic rocks of the Main Series (46.6–49.4 Ma). While ^{40}Ar–^{39}Ar dates will be necessary to confirm the K–Ar dates above, it is possible that these Akobo basalts are the first manifestation of Cenozoic volcanism in East Africa. I opt to include this basalt within the broad Eocene package from 45 to 35 Ma.

This Amaro/Gamo unit, which erupted over a ~10 Ma interval is ~500 m thick and occupies an estimated volume of 30,000 km^3, thus the magmatic flux is ~0.0003 km^3/yr, significantly lower than the subsequent Oligocene flood basalts erupted further north (0.06–0.15 km^3/yr) (Ebinger et al., 1993; George et al., 1998). This early magmatic phase is overlain by the Early Miocene syn-rift Getra Kele basalts, which are separated from the Amaro Tuff by an angular unconformity (Ebinger et al., 1993).

Within the Turkana basin, which lies directly to the south of Southern Ethiopia, detailed stratigraphy over the last 50 years has resulted in well-constrained, though dispersed, stratigraphic columns (Brown and McDougall, 2011). Initial magmatism in this region is recorded at a locality known in the vicinity of Mt. Porr and Kajong on the east side of Lake Turkana at 39.2–35.2 Ma, though the lower and upper temporal bounds of the activity is unconstrained (Furman et al., 2006b; Wilkinson, 1988). The oldest basalt mapped in this region is the Eocene Balesa Koromto unit to which I ascribe this initial phase of magmatism (Fig. 4). The Chibelet Volcanics, which occur on the western shore of Lake Turkana, are considered correlative with the Balesa Koromto on the eastern shore (Ochieng' et al., 1988). A contemporaneous 1000 m sequence of basalts (ca. 35–37 Ma) is evident at Lokitaung, which overlies the Cretaceous Lubur Sandstone and is also known as the Turkana Volcanics (Bellieni et al., 1981; McDougall and Brown, 2009; Morley et al., 1992; Tiercelin et al., 2012).

To the northeast of Lake Turkana, Eocene magmatic activity is observed in the Nabwal Hills (Turkana) and equivalent latitudes across the border in southern Ethiopia. The ~500 m thick Nabwal formation, which is an intercalation of dominantly aphyric transitional basalts and subordinate ignimbrites is the basal member of the ~1400 m thick Asile group, where volcanism continued to ca. 13 Ma. The Nabwal formation is best constrained by a 34.3 Ma ^{40}Ar–^{39}Ar age on feldspar separates from an intercalated ignimbrite 300 m from the base (McDougall and Watkins, 2006), meaning that magmatism extended into the latest Eocene. Across the border in southern Ethiopia, the 150 m thick Fejej basalt, which is dated using K–Ar methods at 34.2–32.8 Ma (Asfaw et al., 1991; Davidson and Rex, 1980) is considered equivalent to the Nabwal basalts. Geochemical investigations have revealed that the incompatible trace element profile of the Fejej basalts are identical the earlier Gamo basalts (Stewart and Rogers, 1996), and the range of radiogenic isotopes is similar. I have thus grouped the Fejej basalt with the contemporaneous Gamo basalts.

2.2. Geochemical characteristics

The Amaro and Gamo basalts have similar trace element and isotopic characteristics however, some important distinctions can be made. The Amaro basalts are most always the more primitive in terms of MgO (Fig. 5), are tholeiitic, and have less incompatible trace element enrichment in comparison to the Gamo basalt (Fig. 6). George and Rogers (2002) suggest a geochemical division between the groups can be made such that the Amaro basalts have $TiO_2 < 2.3$ wt.%, while the Gamo basalts have $TiO_2 > 2.3$ wt.%. The geochemical differences between the groups were ascribed to variations in the depth of fractional crystallization between the suites, such that deep fractionation of clinopyroxene from the Gamo magmas resulted in a shift in magma composition from tholeiitic to alkaline. Little distinction can be made in terms of radiogenic isotope variation between the basalts in $^{87}Sr/^{86}Sr$ (0.7035–0.7045) or $^{206}Pb/^{204}Pb$ (18.1–19.1), however the Amaro basalts are consistently displaced to more radiogenic values of $^{143}Nd/^{144}Nd$ (0.51270–0.51273) in comparison to the Gamo (0.51259–0.51268). George and Rogers (2002) ascribed this difference to the assimilation of a different crustal component as the Gamo basalts fractionated at depth within the lithosphere. Given the acknowledged influence of lithospheric reservoirs on the Amaro and Gamo basalts, it remains difficult to fully resolve the origin of the reservoir which melted to form these basalts.

The thick basaltic piles of Lokitaung and the Nabwal Hills lie at the northern end of the Turkana basin, and are thus closer to the Amaro/Gamo exposures in Southern Ethiopia in comparison to the Balesa Koromto rocks some ~150 km further south. There is limited geochemical data available for the Lokitaung basalts, consisting of major element analyses (Bellieni et al., 1981; Tiercelin et al., 2012), and I am unaware of any existing compositional information on the basalts of the Nabwal Hills. The extant Lokitaung data reveals a fairly wide range in composition (~3.6–7.1 wt.% MgO) that is relatively enriched in TiO_2 (1.8–3.4 wt.%), broadly similar to the temporally equivalent Gamo basalts from Southern Ethiopia. However, CaO/Al_2O_3 is distinctly higher for the Lokitaung basalts, perhaps reflecting a different fractionation assemblage dominated by feldspar, contrasting with the clinopyroxene control within the Gamo basalts. The Balesa Koromto basalts (Kajong/Mt. Porr in Furman et al., 2006b) have been relatively well geochemically characterized. The majority of the Balesa Koromto basalts exhibit a distinctively enriched incompatible trace element pattern with some

Fig. 2. Cenozoic magmatic activity in Africa and parts of the Arabian Peninsula derived from the Geologic Map of Africa (Thiéblemont, 2016). Magmatic activity in East Africa is divided into three periods: Paleogene (66 to 23 Ma); Neogene (23 to 2.6 Ma); Quaternary (2.6 Ma to today). Major volcanic provinces are labeled. Background image is a colored SRTM (NASA) digital elevation model (PIA04965). Major faults, labeled as gray lines, are from Chorowicz (2005).

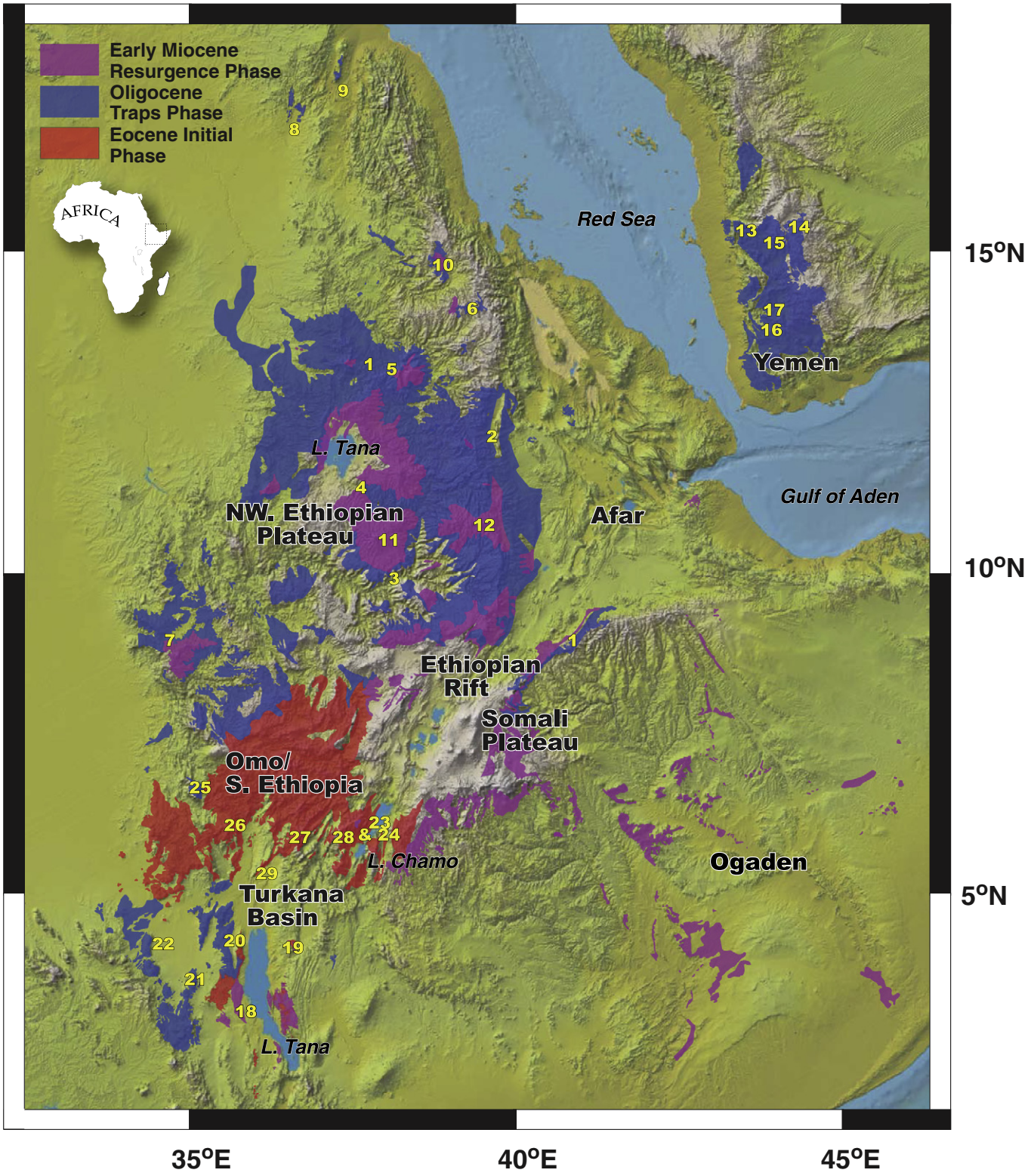


Fig. 3. Magmatic activity in Africa and parts of the Arabian Peninsula from 45 to 22 Ma and divided into three broad periods of magmatic activity: (A) Eocene Initial Phase (45–34 Ma), (B) Oligocene Traps phase (~33.9–27 Ma), (C) Early Miocene resurgence phase (26.9 Ma to 22 Ma). These data are derived from: Yemen (Grolier and Overstreet, 1978; USGS, 1963); Ethiopia (Davidson, 1983; Tefera et al., 1996); Eritrea (Abbate et al., 2014; Teklay et al., 2005); Sudan and S. Sudan (GMRD-Sudan, 1981; Kenea et al., 2001); Djibouti (Le Gall et al., 2015); Somalia (Kassim et al., 1992); Kenya (BEICIP, 1987; Key and Watkins, 1988; Ochieng' et al., 1988; Walsh and Dodson, 1969; Wilkinson, 1988). Background image is a composite image that is comprised of a colored SRTM (NASA) digital elevation model (PIA04965) overlain by elevation derived from GeoMapApp. Note the yellow numbers in this figure refer to stratigraphic sections described in Section 7 and illustrated in Figs. 22–23.

exhibiting a prominent negative K anomaly (Furman et al., 2006b) (Type B — Fig. 7). Furman et al. (2006b) suggest melting of an amphibole-bearing source with perhaps the influence of carbonatite. These basalts have particularly radiogenic $^{206}Pb/^{204}Pb$ (18.75–20.33), with unradiogenic $^{87}Sr/^{86}Sr$ (0.7031–0.7034), and $^{143}Nd/^{144}Nd$ (0.51276–0.51283) slightly more radiogenic than the Amaro basalts.

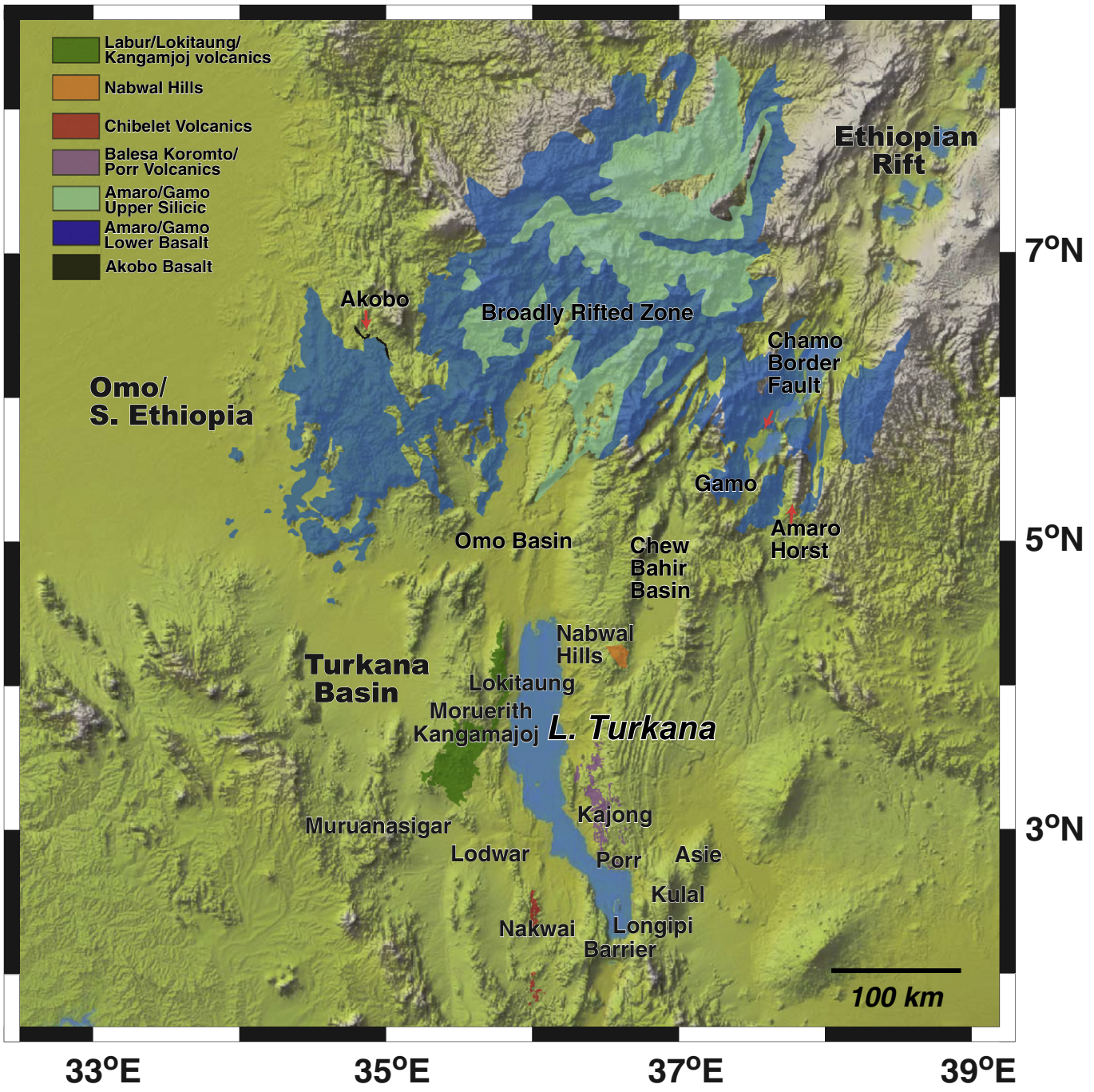


Fig. 4. Magmatic activity in East Africa during the Eocene Initial Phase (45–34 Ma). Data sources the same as in Fig. 3.

This isotopic signature is characteristic of a HIMU-like mantle reservoir (Furman et al., 2006b), though the location of this reservoir within the mantle remains unconstrained. A subset of the Balesa Koromto basalts exhibit a distinctly more depleted trace element signature characterized by low La/Yb and high Zr/Nb (Fig. 5D), a significant positive Ba anomaly, and a primitive mantle normalized pattern which overlaps with the Amaro basalts (Type A: Fig. 7). There is little distinction between types A and B isotopically, though there may be slightly more radiogenic $^{143}Nd/^{144}Nd$ and less radiogenic $^{87}Sr/^{86}Sr$ in the Type B Balesa Koromto basalts.

The extant geochemical data for the Eocene Initial Phase suggests that there are three dominant geochemical signatures in East Africa from 45 to 35 Ma: (1) a variably depleted incompatible trace element signature typified by the Amaro and Gamo basalts, and characterized by a positive Ba anomaly. This signature has less radiogenic $^{206}Pb/^{204}Pb$ and more radiogenic $^{87}Sr/^{86}Sr$ in comparison with the Balesa Koromto basalts. (2) The Balesa Koromto Type A basalts, which are similar to the Amaro/Gamo basalts but can extend to more radiogenic values of $^{206}Pb/^{204}Pb$ and less radiogenic $^{87}Sr/^{86}Sr$. (3) The Balesa Koromto Type B basalts which are more consistently enriched in incompatible trace elements, in particular for elements such as Nb and displaying a prominent negative K anomaly (Fig. 7). The isotopic data are very limited but this magma type may have more consistently radiogenic $^{206}Pb/^{204}Pb$ and less radiogenic $^{87}Sr/^{86}Sr$ than Type A Balesa Koromto basalts.

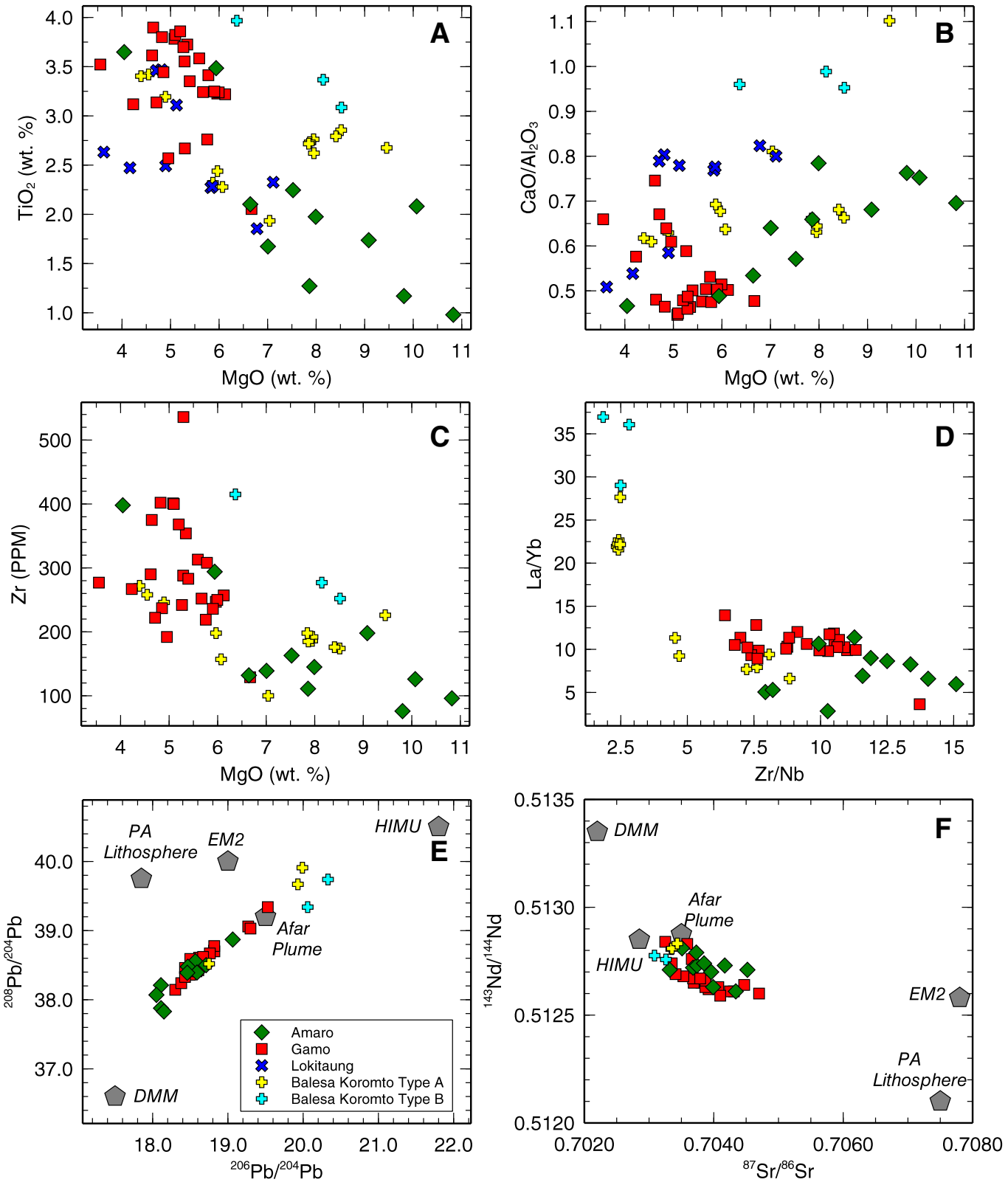


Fig. 5. Geochemical variation of mafic lavas during the Eocene Initial Phase of activity in East Africa/Yemen. Isotopic values are those corrected for eruption age, where reported. The isotopic reservoirs plotted (PA lithosphere — Pan-African Lithosphere; Afar Plume; and DMM — Depleted MORB mantle are from Rooney et al. (2012a). HIMU and EM2 are that of Hart et al. (1992). Data sources: Amaro & Gamo — Stewart and Rogers (1996); George and Rogers (2002): Lokitaung — Bellieni et al. (1981); Tiercelin et al. (2012): Balesa Koromto (Mt. Porr/Kajong) — Furman et al. (2006b). Please note that a unified data table is not presented in this contribution. The interested reader is directed to data published in the citations noted above and collated at GEOROC (http://georoc.mpch-mainz.gwdg.de/georoc/).

2.3. Origin of magmatism during the Eocene Initial Phase

The Turkana basin has the unusual geodynamic characteristic of having been modified during multiple rifting events. The structure of the basin is dominated by the Cenozoic extension associated with the East African Rift as well extension associated with the Cretaceous Anza graben, which crosses northern Kenya at Turkana (Bosworth and Morley, 1994; Ebinger and Ibrahim, 1994). When both rifting events are combined, the resulting stretching factor ($\beta = 1.65$: Hendrie et al., 1994; Mechie et al., 1997) is the most pronounced known within the southern portions the rift. While the Turkana basin is now the zone of significant extension, existing data suggest that the development of half grabens commenced in the Late Oligocene (Morley et al., 1992). However, extension was present in this region throughout the Paleogene (Morley et al., 1999). The Eocene volcanic activity at Turkana is thus syn-extensional, but has been considered related to a mantle thermo-chemical anomaly, herein referred to as a mantle plume (Furman et al., 2006b). While there has been speculation that the northward migration of rifting from Turkana through the Broadly Rifted Zone might have been a causative factor in the generation of Amaro and Gamo magmas (George and Rogers, 2002; Stewart and Rogers, 1996), there is no evidence of rift development here until the Miocene (Ebinger et al., 1993; Pik et al., 2008). Accordingly, the origin of Amaro–Gamo volcanic group is also considered to be related to a mantle plume (George and Rogers, 2002; Pik et al., 2008; Stewart and Rogers, 1996). Such a mantle plume origin is consistent with significantly elevated mantle potential temperatures recorded for the Amaro Basalts (1500 °C: Rooney et al., 2012b), however the precise geodynamic context of this plume and its relationship to other plumes in East Africa (e.g. Turkana) remains controversial and will be addressed in Section 6. As the current understanding of the origin of Eocene magmatism in the Turkana/Southern Ethiopia region is unclear, it is germane to briefly outline the proposed models here. There are broadly four models which describe the potential context of the plume in this region: (A) There is a single plume throughout East Africa (Ebinger and Sleep, 1998), (B) There is a single plume at depth but with distinctive composition heads influencing Tanzania and Afar (Furman et al., 2006b), (C) Convective instabilities associated with this single plume generated magmatism in advance of the Oligocene flood basalt event (Pik et al., 1999), (D) There are discrete Kenyan and Ethiopian mantle plumes and that Southern Ethiopia was influenced by the Kenya plume (George and Rogers, 2002).

3. Oligocene Traps phase

3.1. Southern Ethiopia/Turkana

Here, I define the Early Oligocene Traps phase as magmatism subsequent to the ~34 Ma termination of the Gamo and Lokitaung basalts and continuing through to the termination of significant silicic volcanism following the Ethiopian–Arabian flood basalts at ca. 27 Ma. This period, unlike the relatively restricted magmatism of the Eocene, is widespread, and voluminous (Fig. 8). I begin our narrative where we left off — in Southern Ethiopia and the Turkana Basin, and subsequently expand to northern Ethiopia and Yemen.

Early Oligocene activity is evident within the Turkana basin during this period (Brown and Jicha, 2016). To the east of the Nabwal Hills is the isolated ~120 m thick basalt (El Yibo basalt) whose age is poorly constrained but is considered likely Oligocene–Early Miocene. Isolated occurrences of ca. 33 Ma basalts are evident in the Nakoret Hills and Muruanchok Hills (Morley et al., 1992; Zanettin et al., 1983), however more significant occurrences are evident in the western part of the Turkana basin, specifically: (A) along the western edge of the Turkana basin, and (B) within the Lokitaung region. Within the Mogila and Songot ranges, which lie along the western edge of the Turkana basin, 1800 m of basalt has been noted, but the precise age of the basalt portion of the sequence is poorly constrained (Wescott et al., 1999). Wescott et al. (1999) suggest the basalt is correlative with those from the Muranchok Hills (~100 km to the southwest) and is thus ~33 Ma on the basis of a K–Ar date by Zanettin et al. (1983). The only robust constraint of the age of rocks from this area is a high precision feldspar separates ^{40}Ar–^{39}Ar date from the 900 m of conformably overlying rhyolites, dated to 32.49–31.67 Ma (Brown and Jicha, 2016). In the regions surrounding Lokitaung, ~1000 m of silicic magmatism from the Kachoda valley is dated via K–Ar to ca. 31.1–32.3 Ma (McDougall and Brown, 2009) and ~1500 m of basalt from the Kalin/Moruerith volcanic sequence also dated by K–Ar to 29.3–31.5 Ma (Bellieni et al., 1981; Zanettin et al., 1983).

While the exact volumes erupted are unconstrained, seismic evidence suggests that the early Oligocene volcanic sequences are continuous beneath the Lotikipi plain (Wescott et al., 1999), and therefore represent a substantial volume that underlies much of the western Turkana basin (ca. 24,000 km^3 of basalt on the basis of the dimensions of the basin). Very limited geochemical data exist for the early Oligocene basalts in Turkana, however major element XRF data suggest variability from silica under-saturated (normative nepheline) to more transitional (normative hypersthene) (Bellieni et al., 1981), distinctly different from the normative quartz values calculated for the underlying ~35 Ma Lokitaung basalts.

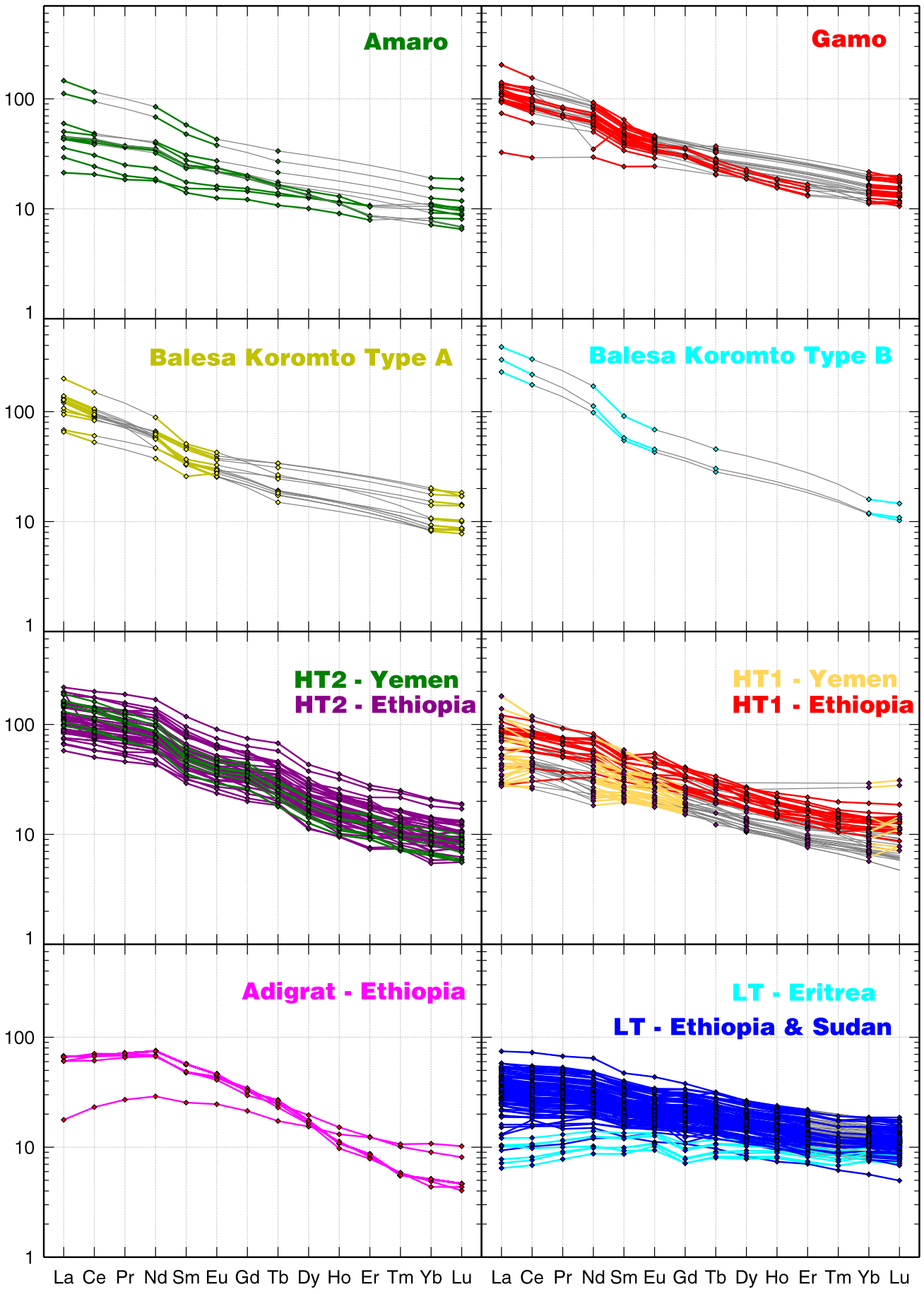
In Southern Ethiopia, at a longitude similar to the Lotikipi Plain in the western Turkana basin (35°E), the Makonnen basalt (formerly subdivided as the Gok and Makonnen basalt by Davidson and Rex, 1980) occupies a maximum thickness of 700 m and has been dated using K–Ar methods from 28.8 to 34.8 Ma, with a majority of dates clustering at 31–32 Ma (Davidson, 1983). The dominantly columnar flows of the Makonnen basalt lie dominantly above crystalline basement but is, in places, in contact with the underlying Eocene Amaro/Gamo basalts separated by a gentle unconformity (1.7°: Davidson, 1983). There is a notable absence of silicic volcanism within the Makonnen basalts sequence. Given the similar ages, the corresponding latitudes and thicknesses of basalts we tentatively suggest that the Makonnen Basalts are equivalent to the Early Oligocene basalts in the western Turkana basin discussed above (Fig. 8). Significantly more geochemical data is needed to test this hypothesis, however it is becoming clear that a significant volume of early Oligocene basalts erupted west of Lake Turkana, temporally correlative with the Ethiopian–Arabian flood basalts to the north.

3.2. Flood basalts on the Ethiopian Plateau

3.2.1. Historical evolution of the nomenclature of the Ethiopian Traps

For the past 150 years, the study of the Ethiopian–Arabian flood basalts has been biased, for anthropologic reasons, towards the northwest Ethiopian Plateau. The location of the former Ethiopian capital city in Gondar atop the flood basalt sequence resulted in initial maps of this region in the 1860s and 1870s (Mohr, 2009). Blanford, during this time period was the first to use the term 'Ashangi' to describe the flood-basalt sequence (Blanford, 1869), wherein it was defined as the basal basaltic unit of the flood basalts, separated by an angular unconformity from the overlying basaltic and trachytic Magdala group (Mohr, 2009). Continued investigations over the next 100 years resulted in the adoption of a four-fold division of the 'trap series' by some authors from oldest to youngest (e.g., Zanettin et al., 1980):

1. Ashangi basalt. This was defined as being the lowest basalt on the northwest Ethiopian Plateau and was characterized by deeply altered flood basalt that was tilted up to 40° (Mohr and Zanettin, 1988). This member of the flood basalt group was limited to central Ethiopia.
2. Aiba fissural basalt. This transitional flood basalt unit unconformably overlies the Ashangi basalt, erupting onto the Ashangi peneplain. The flows are 15–50 m thick and may be columnar (Mohr and Zanettin, 1988).


Amaro
Gamo
Balesa Koromto Type A
Balesa Koromto Type B
HT2 - Yemen
HT2 - Ethiopia
HT1 - Yemen
HT1 - Ethiopia
Adigrat - Ethiopia
LT - Eritrea
LT - Ethiopia & Sudan
100
10
1
La Ce Pr Nd Sm Eu Gd Tb Dy Ho Er Tm Yb Lu

3. Alaji fissural basalts and rhyolite. This is the final stage of the widespread flood basalt sequences and is intercalated with rhyolite horizons that are tens of meters thick (Mohr and Zanettin, 1988).
4. Termaber basalts. These are more localized basalts which overlie the Alaji unit in places. They are derived from large (tens of kilometers) low-angled shields but are fissure fed (Mohr and Zanettin, 1988)

While the relative stratigraphic order was constrained through the principle of superposition, defining the precise ages of these sequences was fraught with difficulty given the imprecision of the K–Ar technique on altered basaltic materials (e.g., Zanettin et al., 1980). Thus the divisions were made on the basis of lithologic or structural discontinuities between units. The unconformity between the Ashangi and Aiba basalts became an important marker, however this unconformity is suggested to be localized to the original type section of the Ashangi basalts, and is not seen elsewhere on the plateau (Mohr, 1983). The division between the Aiba and Alaji was defined as the first appearance of an ignimbrite bed, however these ignimbrite beds are not always present, making it difficult to distinguish between these two units in the absence an ignimbrite (Mohr, 1983). Furthermore, the potentially wide age variation in the Alaji and Termaber made it difficult to establish a clear chronology (Mohr and Zanettin, 1988). These difficulties resulted in significant confusion on the precise stratigraphy of the Ethiopian Trap series in papers published prior to 1996.

These stratigraphic nomenclature issues are readily apparent when applied to the poorly exposed successions in the central Ethiopian Plateau. Here, the stratigraphic work undertaken in the south of Ethiopia, which resulted in the recognition of the Makonnen and underlying undivided succession (Davidson, 1983), meets the four-fold division established in the north of the plateau (Zanettin et al., 1980). Country-scale maps show the early Oligocene Makonnen basalts extending to 10°N, curiously terminating on the south bank of the Abay (Blue Nile) River (Fig. 8), and also on the Somali Plateau to the East (Tefera et al., 1996). Exposures to the north of the Abay River, are then mapped on the basis of the four-fold division noted above. This dichotomy has resulted in a conceptual disconnection between magmatic events in the Ethiopia and Turkana during the Eocene and early Miocene.

A different stratigraphic nomenclature developed for the flood basalts in Eritrea and Sudan. In Eritrea, Zanettin et al. (1999) proposed a lower 200 m thick Asmara basalt (locally the Menguda basalt: Teklay et al., 2005), with an undefined relationship to another group termed Alaji. Overlying the Alaji is the Adi Ugri (locally Ona and Durko basalt: Teklay et al., 2005), which was characterized as basalt flows that were intercalated with laterites (Zanettin et al., 1999). Subsequent ^{40}Ar–^{39}Ar dating constrain the Alaji and Asmara basalts to being prior to 29.03 Ma and thus part of the flood basalt series, in contrast the Adi Urgi basalts are post-23.55 Ma (Abbate et al., 2014) and are unrelated to the Early Oligocene magmatic episode.

Further north, along the Eritrea/Sudan border, the Derudeb basalts, which were deposited into two extending basins, have reliable ^{40}Ar–^{39}Ar dates on feldspar separates from the upper portion of the column, and return values of 29.9 Ma. A common overlying rhyolite is dated to 29.6 Ma (Kenea et al., 2001). Further exposes of the flood basalts are evident at Gedaref, along the Ethiopia-Sudan border (Lucassen et al., 2008).

3.2.2. Modern interpretations of the Ethiopian flood basalts

With the application of new ^{40}Ar–^{39}Ar techniques and geochemical characterization of the trap series basalts, it became apparent that the source of ambiguity with the previous division of the trap basalts resulted from:

(A) a compressed period of eruption for the entire traps in Ethiopia, Yemen, and Eritrea (ca. 31–29 Ma: Baker et al., 1996a; Hofmann et al., 1997; Abbate et al., 2014), which was not resolvable with previous K–Ar techniques. ^{40}Ar–^{39}Ar geochronology significantly constrained the possible period during which the flood basalt episode could have occurred, but some degree of ambiguity remains. In Ethiopia, the lowest flood basalt was dated at 31.2 Ma (Hofmann et al., 1997; Ukstins et al., 2002), however, the 2σ error recalculated by Ukstins et al. (2002) for this sample (1.2 Ma) permits the initial flood basalt phase to have begun earlier. Ukstins et al. (2002) provide a date of 30.9 Ma for the Ethiopian flood basalts (±0.11) but the sample is not from the base of the sequence, again permitting an earlier start to volcanism. A rhyolite from near lake Tana was dated using zircon at 31.11 Ma (±0.04 Ma) (Prave et al., 2016) but is underlain by basalt, again permitting an earlier date. In contrast, the sequence in Yemen is better constrained and could not have commenced prior to 30.9 Ma (±0.24 Ma) (Baker et al., 1996a).

(B) A profound spatial variation in magma composition where low-titanium type flood basalts were favored on the periphery of the northwest Ethiopian Plateau, while high-titanium flood basalts occurred towards the center of the traps created spatial heterogeneity in the trap series (Pik et al., 1998, 1999). The profound lateral variation in composition of the Ethiopian Trap lavas (Fig. 8), and the apparent synchronicity in eruption, make the prior plateau-wide division into the Ashangi, Aiba, and Alaji groups unhelpful in modern magmatic studies (Kieffer et al., 2004; Pik et al., 1998, 1999). Instead I subdivide my discussion into low and high-Ti (HT) compositional domains. The low-Ti (LT) province lying west of a line between Adrigrat and Debre Markos, the high-Ti basalts lying to the east (Fig. 8). Beccaluva et al. (2009) and Natali et al. (2016) extended this geochemical division to Yemen, where HT2 flood basalts are reported around Sana'a and HT1 basalts elsewhere (Fig. 8). It should be noted that the spatial divisions are broad and that LT flood basalts can be seen to make up the lower portion of the Oligocene Traps in contact with the underlying basement, even if the main eruptive volume is HT1 (Kieffer et al., 2004).

3.2.3. High-Ti flood basalts

The high-Ti (HT) flood basalts of northwestern Ethiopia form a distinctive geochemical domain that is characterized by transitional basalts with Ti/Y > 450 and Nb/Y > 0.45 (Pik et al., 1998). These HT basalts are further sub-divided into HT1 (relatively lower Ti/Y and Nb/Y) and a HT2 group with generally higher values of these ratios (Ti/Y > 700 and Nb/Y > 0.90). HT1 lavas commonly occupy a transitional compositional space between HT2 and low Ti (LT) basalts in terms of minor and incompatible element enrichment (e.g., Zr, TiO_2, Zr/Nb; Fig. 10). HT2 basalts are dominantly found in the eastern portion of the northwestern plateau, closer to the modern rift valley and the city of Lalibela (Fig. 8: Beccaluva et al., 2009; Pik et al., 1998).

The petrographic characteristics of the HT1 group are quite variable and as in other series, the textures vary depending on locality and stratigraphic position (Pik et al., 1998). The only commonality identified is the presence of purple clinopyroxene and K-feldspar — a trait shared with the HT2 group (Pik et al., 1998), though the HT1 group is considered sparsely porphyritic (Beccaluva et al., 2009). From a geochemical perspective, HT1 basalts typically have incompatible trace element profiles that are quite variable (Figs. 6, 11), and are more evolved (Mg# < 59) than other flood basalt lava series (Beccaluva et al., 2009; Pik et al., 1998). The HT1 group has variable isotopic values that extend

Fig. 6. Chondrite normalized data (Boynton, 1984) of mafic lavas erupted during the Eocene Initial Phase and Oligocene Main Phase of activity in East Africa/Yemen. Data sources are the same as Figs. 5 and 10.

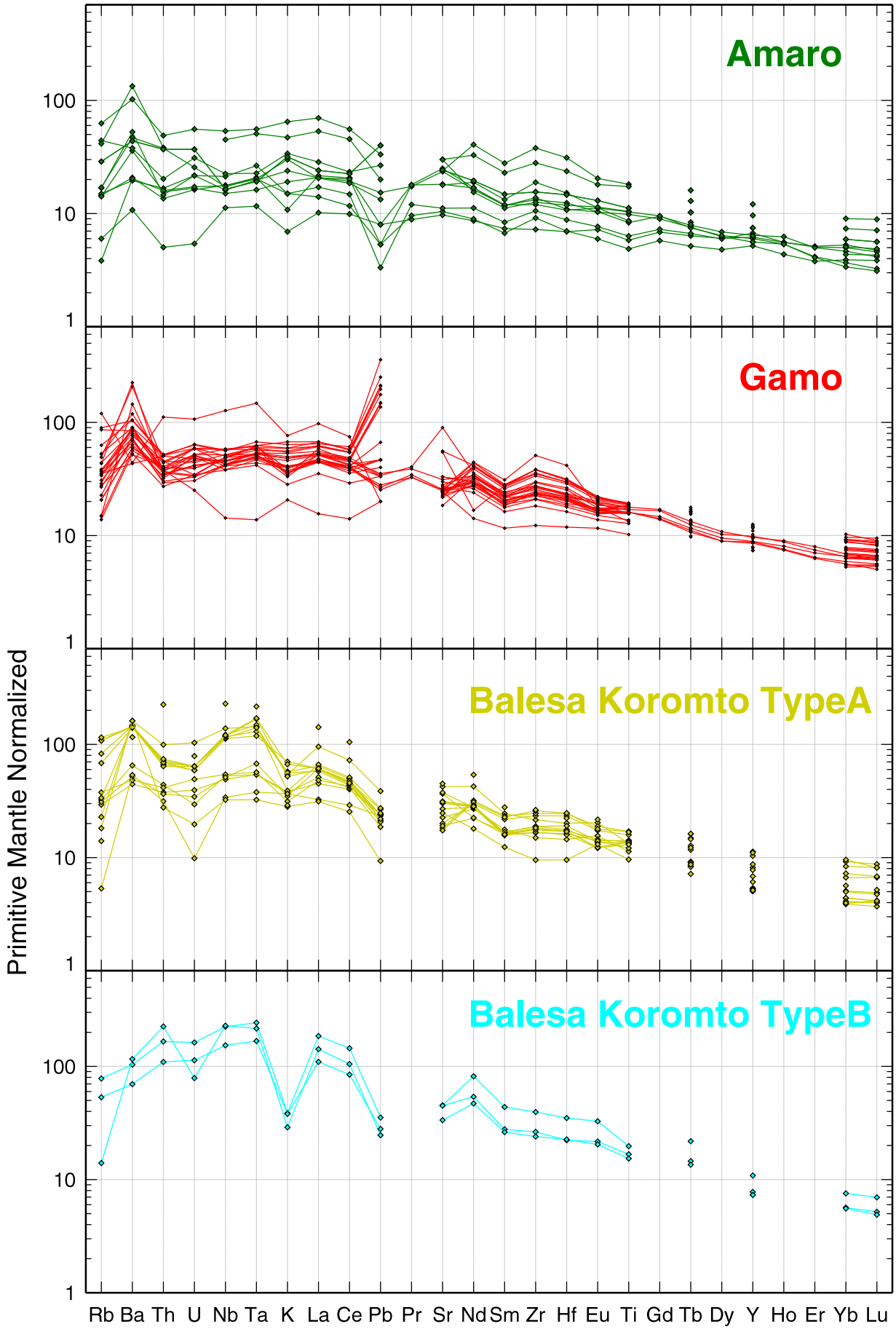


Fig. 7. Primitive mantle normalized data (Rooney et al., 2017) of mafic lavas erupted during the Eocene Initial Phase of activity in East Africa/Yemen. Data sources are the same as Fig. 5.

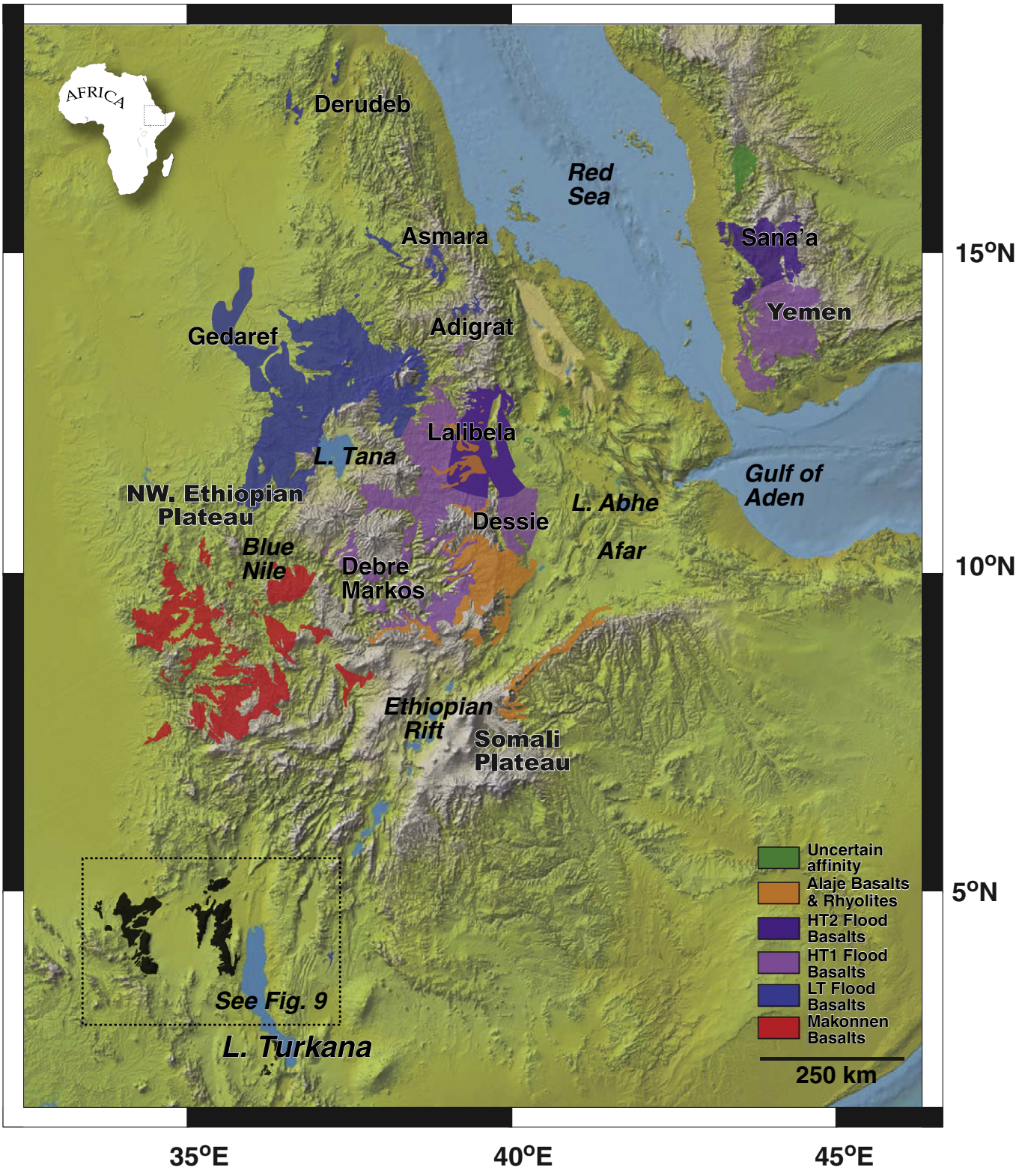


Fig. 8. Magmatic activity in East Africa/Yemen during the Oligocene Traps Phase (33.9–27 Ma). Data sources the same as in Fig. 3. The inset box refer to Fig. 9.

from $^{206}Pb/^{204}Pb$ of 18.46 to 18.79, and $^{87}Sr/^{86}Sr$ ~ 0.7035 to 0.7043 (Fig. 10).

The HT2 group is characterized by being more porphyritic than other groups, and can be impacted by the accumulation of olivine and pyroxene phenocrysts. Phlogopite and alkali feldspar are seen but are rare (Beccaluva et al., 2009). Crystal accumulation within the HT2 group results in elevated Mg# up to 83 and Ni contents (up to 1373 ppm: Beccaluva et al., 2009), and is clearly unreflective of a

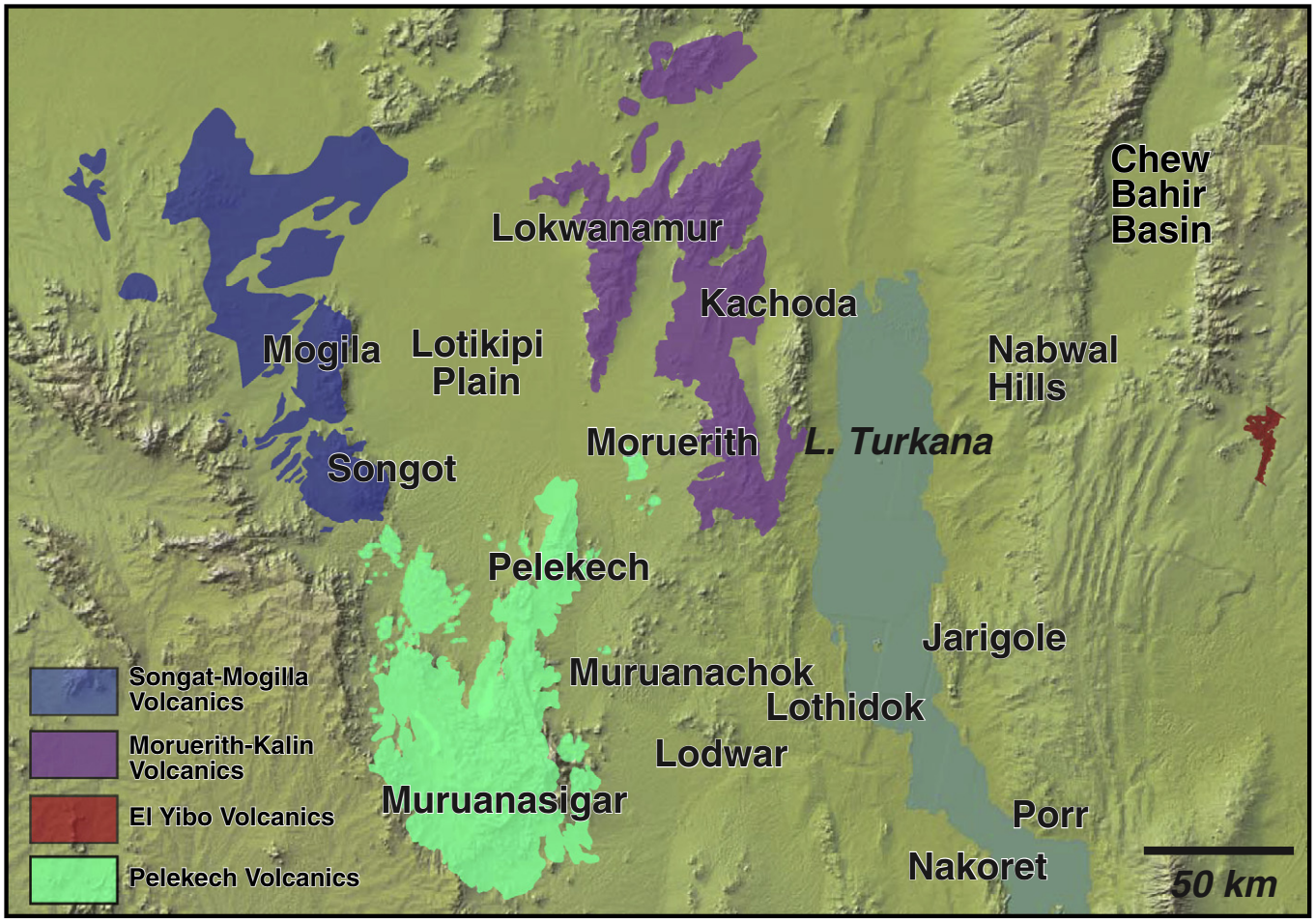


Fig. 9. Magmatic activity in the Turkana Basin during the Oligocene Traps Phase (33.9–27 Ma). Data sources the same as in Fig. 3.

primitive magma. Beccaluva et al. (2009) suggest picrites do occur without cumulus olivine and have a Mg# of 68–72. In contrast to the HT1 group, the incompatible trace element profile of the HT2 group is quite distinctive, characterized by a strong negative K anomaly (Fig. 11), similar to some Eocene magmas from the southern portions of the Turkana depression (Balesa Koromto Type B). The HT2 group exhibits some of the most radiogenic $^{206}Pb/^{204}Pb$ of up to 18.86, and has been suggested to best represent the isotopic composition of the Afar plume (Pik et al., 1999).

3.2.4. Low-Ti (LT) flood basalts

The LT flood basalts occupy the western portion of the northwestern Ethiopian Plateau and continue northward into Eritrea (Beccaluva et al., 2009; Kieffer et al., 2004; Teklay et al., 2005), and westward into Sudan (Kenea et al., 2001; Lucassen et al., 2008). The LT flood basalts are the predominant flood basalt group by volume (Beccaluva et al., 2009). The LT group is distinctive from the two HT flood basalt types in that it follows a tholeiitic crystallization sequence (Beccaluva et al., 2009). Olivine phenocrysts are generally sparser in comparison to the HT groups, and plagioclase is a more common phase. LT basalts exhibit sub-ophitic texture within their groundmass (plagioclase laths with interstitial clinopyroxene). The LT basalts plot within the subalkaline field and show a wide range in compositions from Mg# of 36 to 68 (Beccaluva et al., 2009; Kieffer et al., 2004; Pik et al., 1998), though some units from south of Asmara (Eritrea) have particularly elevated Mg# that can be as high as 75, suggesting accumulation of olivine in a similar manner to the HT2 lavas (Teklay et al., 2005). Other than low TiO_2 (Fig. 10), a distinctive difference between the LT and HT flood basalt groups is the broad depletion in incompatible elements displayed by the LT flood basalts (Fig. 6; 11). The LT series exhibit characteristic positive Ba and Sr anomalies, a depletion in HFSE, and broadly exhibit a more coherent trace element signature (Fig. 11). The LT flood basalts isotopically overlap the HT series (Kieffer et al., 2004; Pik et al., 1999; Teklay et al., 2005), but extend to the lowest $^{206}Pb/^{204}Pb$ values of 18.00–18.67, $^{87}Sr/^{86}Sr$ also extends to lower values of 0.7034–0.7044. A notable exception to these unradiogenic Pb isotope values is observed at the margins of the LT flood basalt province at Gedaref in Sudan (Lucassen et al., 2008), and in LT dikes from along the Afar margin near Dessie (within the HT1 province) that are dated post-traps ($^{40}Ar–^{39}Ar$ — 27.34 Ma) (Rooney et al., 2013). These samples, while conforming with the other geochemical characteristics of the LT flood basalts, exhibit particularly elevated values of $^{206}Pb/^{204}Pb$ of up to 20.29 (Fig. 10). For all the flood basalts $^{143}Nd/^{144}Nd$ is not particularly diagnostic in terms of classification: The LT flood basalts have 0.5127–0.5130, while the HT series overlap with 0.5128–0.5130, though the HT2 samples consistently exhibit the most radiogenic values (Fig. 10).

3.2.5. Oligocene shield volcanoes, rhyolites, and the termination of the Oligocene event on the Ethiopian Plateau

Towards the termination of the widespread flat-lying trap series on the Ethiopian Plateau, a new period of volcanism was dominated by the construction of the massive Simien shield volcano on top of the existing flood basalt pile, and the sub-ordinate but widespread appearance of silicic volcanism (Ayalew et al., 2002; Kieffer et al., 2004). The flows which constitute the Simien shield dip 4–6° away from the summit

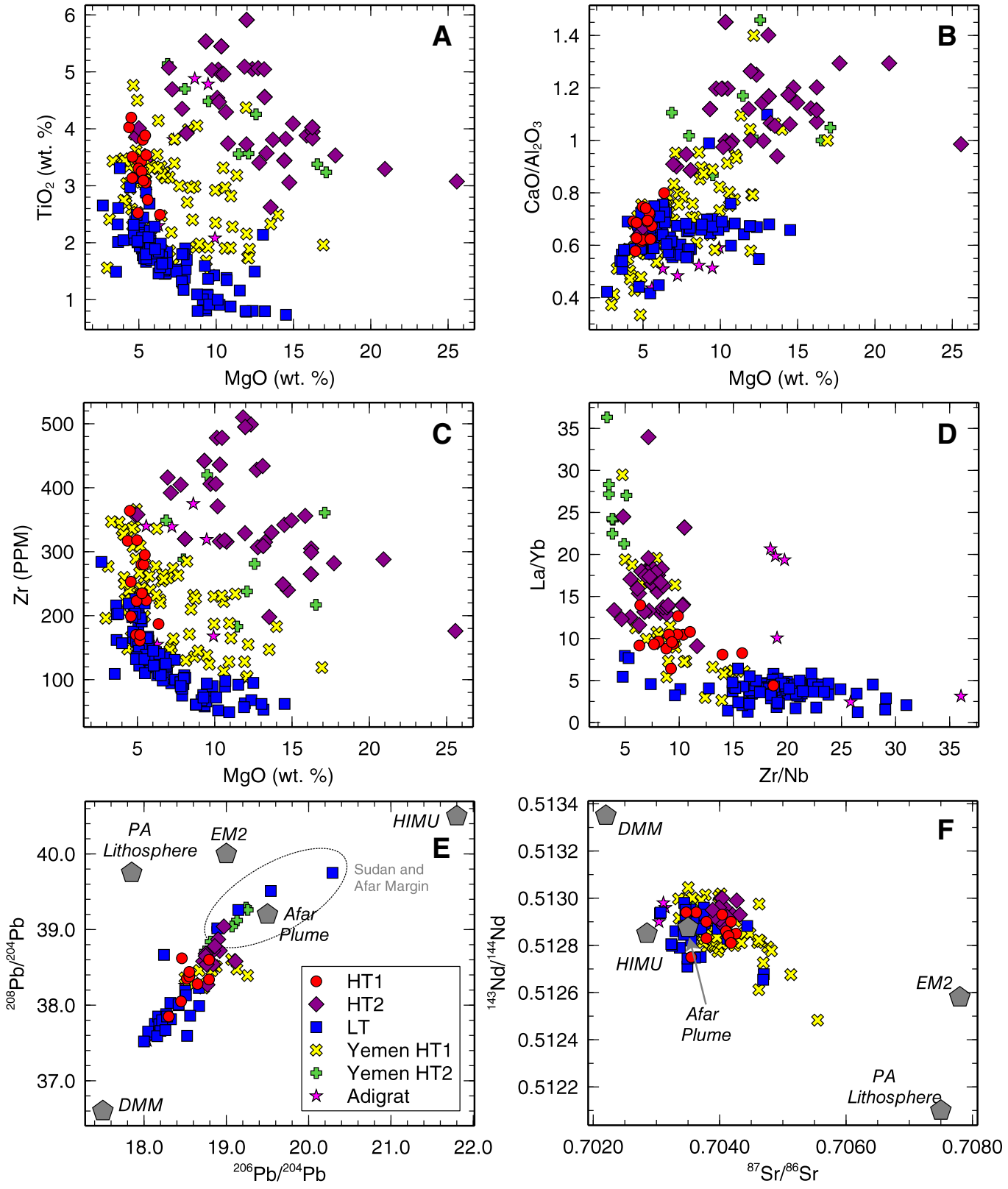
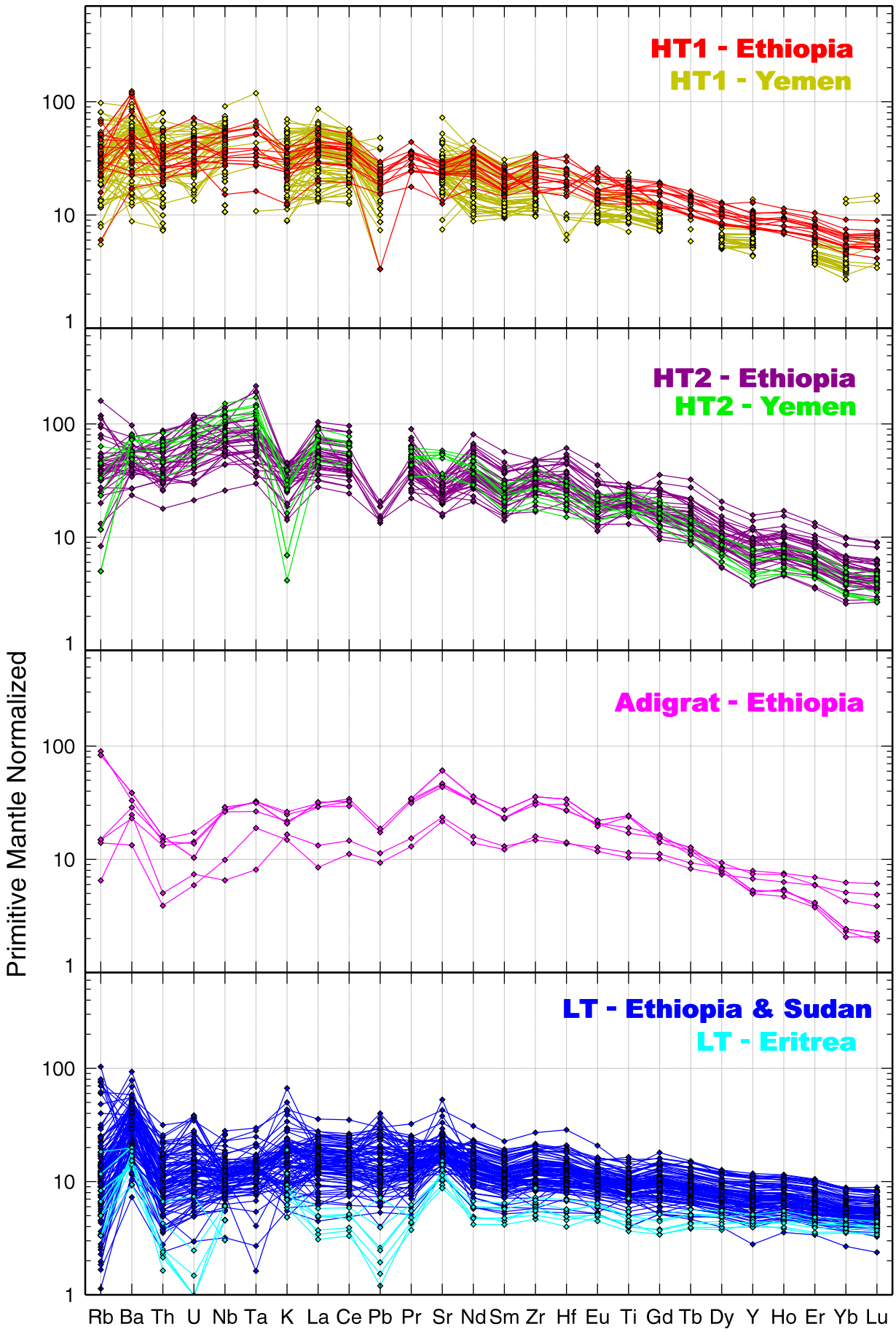


Fig. 10. Geochemical variation of mafic lavas during the Oligocene Traps phase in East Africa/Yemen. Isotopic values are those corrected for eruption age, where reported. The isotopic reservoirs plotted (PA lithosphere — Pan-African Lithosphere; Afar Plume; and DMM — Depleted MORB mantle are from Rooney et al. (2012a). HIMU and EM2 are that of Hart et al. (1992). Data sources: HT1 — Pik et al. (1999); Beccaluva et al. (2009): HT2 — Pik et al. (1999); Beccaluva et al. (2009); Natali et al. (2016): LT — Kieffer et al. (2004); Teklay et al. (2005); Rooney et al. (2013); Lucassen et al. (2008); Pik et al. (1999); Beccaluva et al. (2009): Adrigrat — Pik et al. (1999); Beccaluva et al. (2009): Yemen HT1 — Baker et al. (1996b); Chazot and Bertrand (1993); Manetti et al. (1991); Yemen HT2 — Natali et al. (2016). Note that LT basalts from the margins of the province (Gedaref, Sudan) and as dikes within the proto-rift border faults along the Afar margin are circled in Fig. E denoting as consistently elevated Pb isotope values for these samples.


HT1 - Ethiopia
HT1 - Yemen
HT2 - Ethiopia
HT2 - Yemen
Adigrat - Ethiopia
LT - Ethiopia & Sudan
LT - Eritrea
Primitive Mantle Normalized
100
10
1
Rb Ba Th U Nb Ta K La Ce Pb Pr Sr Nd Sm Zr Hf Eu Ti Gd Tb Dy Y Ho Er Yb Lu

and are separated from the underlying flood basalts with a breccia layer, and are themselves distinctive by being porphyritic (Kieffer et al., 2004). This dip contrasts sharply with the mostly flat lying flood basalts, however the composition of the Simien shield (LT) is similar to the underlying flood basalts suggesting continuity in terms of magma source (Kieffer et al., 2004). I have grouped these lavas with the rest of the LT lavas in geochemical plots. K–Ar and ^{40}Ar–^{39}Ar dates of the Simien shield are consistent with the terminal stages of the flood basalt episode ranging from 29.1 to 29.9 Ma (Coulie et al., 2003; Rochette et al., 1998), further supporting the concept of a continuity in the magmatic system which formed the Simien shield from the underlying flood basalts. Additionally, Kieffer et al. (2004) suggest that a waning magmatic flux in the upper part of the flood basalts is linked with the construction of the Simien shield volcano.

Consistent with a narrative of a waning flux towards the upper portion of the flood basalt sequence, a series of tuffs, ignimbrites, and obsidian up to 500 m thick are found interbedded with the basalts (Ayalew and Yirgu, 2003; Ayalew et al., 2002). Ayalew et al. (2002) suggest a volume of 6.3×10^4 km^3 of silicic materials erupted on the Ethiopian Plateau starting at about 30.2 Ma continuing in more limited volume to 25.3 Ma (Ukstins et al., 2002), consistent with the waning phases of the flood basalt sequence. The majority of these silicic lavas are either metaluminous or peraluminous, though the Wengel Tena lavas (coincident with HT basalts) extend to peralkaline composition (Fig. 12). Temperature estimates of these lavas are between 850 and 950 °C at the QFM buffer (Ayalew et al., 2002; Natali et al., 2011). Similar to the underlying flood basalts, the rhyolites broadly exhibit trends in TiO_2 consistent with a linkage between the rhyolites and basalts (Fig. 12). For example, the lavas at Lima Limo and Lake Tana were erupted over LT flood basalts, these rhyolites are offset to lower values of TiO_2 in comparison to the Wengel Tena and Yemen rhyolites, which were emplaced over HT lavas (Fig. 12). This separation is also evident in elements such as Sr, where rhyolites erupted in the LT and HT provinces follow different evolutionary paths (Fig. 12). Broadly, rhyolites erupted in the HT regions tend to exhibit very similar chondrite-normalized rare earth element (Fig. 13) and primitive-mantle normalized trace element patterns (Fig. 14). There are subtle variations, however, whereby the most incompatible elements (e.g. Nb) tend to be more depleted in the Lima Limo (LT) rhyolites, demonstrated in Nb/La, which separates HT lavas from LT (Fig. 12).

The lack of strong crustal signatures in terms of radiogenic (Fig. 12: $^{87}Sr/^{86}Sr$ 0.7050; $^{143}Nd/^{144}Nd$ 0.5128) and stable isotope (δO^{18} 6.5) values for many of these rhyolites in Ethiopia (Ayalew et al., 2002) is evidence for their derivation from fractional crystallization of basalts with limited to no crustal assimilation — a model consistent with evidence of cumulophyric aggregates discovered in some of the Oligocene ignimbrites (Barbey et al., 2005). Importantly, these data do not support a model of large-scale crustal anatexis or partial melting of an underplate (Ayalew et al., 2002; Natali et al., 2011). Despite evidence of dominantly fractional crystallization processes, rhyolites from Lima Limo near the Simien shield do exhibit $^{143}Nd/^{144}Nd$ lower than the underlying flood basalts (0.5125) suggesting more significant crustal assimilation at this location (Ayalew et al., 2002). Similar to the flood basalts in the northern portion of the Ethiopian flood basalt province, basalts in the Jima area of the central Ethiopian Plateau are overlain by ~700 m of silicic volcanic rocks, which isotopically resemble those to the north and have a Rb–Sr isochron age of ca. 30.5 Ma (Ayalew and Yirgu, 2003). This parallelism suggests a broad waning of flux across the plateau during this time period.

The fractional crystallization model for the formation of the rhyolites on the Ethiopian Plateau has the potential to explain the absence of intermediate composition rocks on the Ethiopian Plateau (the so-called Daly Gap) and compositional linkages between the rhyolites and basalts (Natali et al., 2011). Modeling by Natali et al. (2011) showed that the period of fractionation of Fe–Ti oxides, which increases the SiO_2 content of the residual liquids, has a narrow thermal window (Natali et al., 2011). Furthermore, geochemical modeling shows that the composition of the underlying flood basalts impacts the subsequent composition of the rhyolites (Fig. 12) — for example, the rhyolites overlying the HT2 flood basalts have elevated TiO_2 (0.34 to 0.74 wt.% TiO_2 at 75 wt.% SiO_2; Natali et al., 2011) in comparison to those over the LT flood basalts (0.30 wt.% TiO_2 at 75 wt.% SiO_2) (Ayalew and Yirgu, 2003).

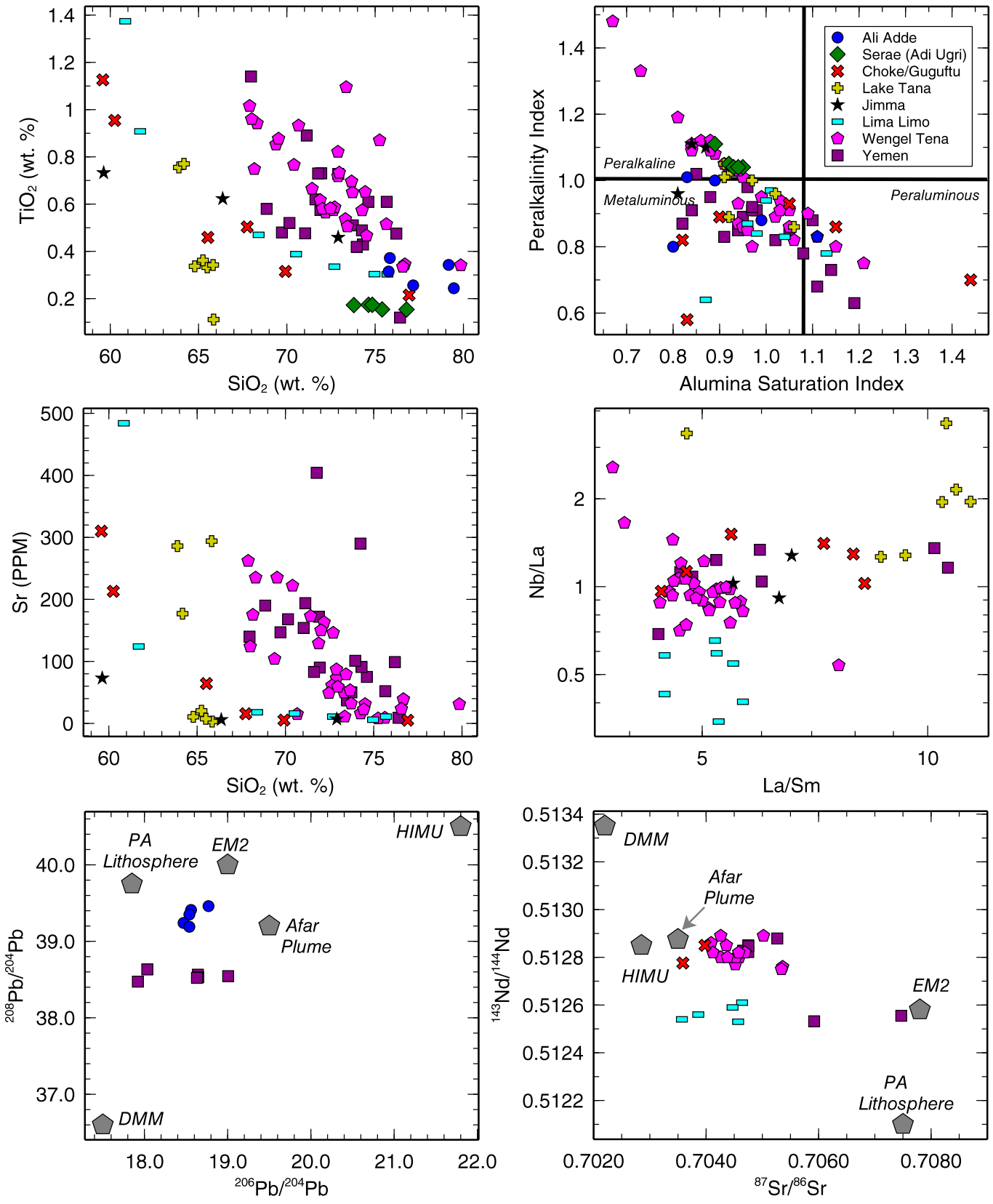
If the Oligocene rhyolites are derived from the same magmatic plumbing system as the preceding basalts, then a natural question arises - what might have changed in the magmatic system to allow for the shallow ponding of magmas? Existing studies point to tectonic rationales and suggest that rifting and block tilting created accommodation space (Natali et al., 2011). It is, however, unclear that this model could account for the appearance of these magmas, as significant upper crustal extension along the Afar margin is thought to be Miocene in age (Wolfenden et al., 2005). Equally, the appearance of rhyolites distant from the rift margin (e.g., Lima Limo), further complicate this model. Other authors have suggested a radial extension model (Ayalew et al., 2002), however to fully evaluate the shift in the magmatic system, a temporal study examining the change in the flood basalt flux and magmatic evolution will be necessary.

3.3. Flood basalts on the Somali Plateau

Most figures which illustrate the extent of the trap series in Ethiopia include the Somali Plateau in the region impacted by the flood basalt episode (Mohr and Zanettin, 1988). With the advent of new high-precision ^{40}Ar–^{39}Ar dating from the northwest Ethiopian Plateau that showed the traps series were constrained to between ca. 31–29 Ma (Hofmann et al., 1997), a logical assumption might be that the trap rocks from the Somali Plateau would be of a similar age. However, little modern ^{40}Ar–^{39}Ar information is available on the trap series from the Somali Plateau. Abbate and Sagri (1980) presented K–Ar data suggesting that the commencement of activity on this plateau is later than on the Adjacent Ethiopian Plateau at ca. 30 Ma, though the majority of dates appear to be younger. Data from Jones (1976) shows K–Ar dates that range from ca. 20 to 24 Ma for the lower flows on the northeast part of the plateau. These earlier studies did deduce some differences between the two plateaus but more precise geochronology was required.

The best-studied portion of the plateau is along the Asebe Teferi escarpment with Afar. Here, between 700 and 3000 m of late Oligocene and early Miocene basalts (35–24 Ma), are overlain by another 800 m of Mid-Miocene flows (18–13.5 Ma) and are collectively termed 'UTB' (Juch, 1978). This amalgamation has been resolved in subsequent studies by Nelson et al. (2017), which show that the lower flows from Asebe Teferi, along the Afar margin, return an ^{40}Ar–^{39}Ar date of ~28.6 Ma, while rocks higher in the stratigraphy are 18.5–15.5 Ma. This division in ages is consistent with observations by Zanettin et al. (1980) and Mohr and Zanettin (1988) that the lower portion of flows along this margin is composed of extensive flows of fine grained transitional basalt (i.e., resembling the Oligocene flood basalts), while the younger flows were distinctly more porphyritic and alkaline. Away from the Asebe Teferi margin, the stratigraphy defined by Juch becomes less robust and thins towards the plateau southward (Mohr and Zanettin, 1988). Early Oligocene dates are, however, found distally as isolated occurrences on the Ogaden plain (28.1–30.0 Ma: Mège et al., 2016). Samples from the lower part of the Asebe Teferi section have geochemical affinities to the HT1 group of the northwest Ethiopian

Fig. 11. Primitive mantle normalized data (Rooney et al., 2017) of mafic lavas erupted during the Oligocene Traps phase in East Africa/Yemen. Data sources are the same as Fig. 10. Note that lavas from Eritrea have been highlighted in cyan due to their distinction from other LT flood basalts. Further work is needed on this group to assess if this is a clear differential between the lava types given the analytical challenges of measuring Pb and U at these low levels.



Plateau (Nelson, 2009; Nelson et al., 2017) including similar isotopic values of $^{206}Pb/^{204}Pb$ (18.5–18.9). Similarly, Oligocene samples from Ogaden are also classified as HT1 (Mège et al., 2016). These Oligocene flood basalts are not, however, the dominant magmatic episode on the Somali Plateau.

Studies by Nelson et al. (2017) and Mège et al. (2016) begin to clarify magmatism on the Somali Plateau away from Asebe Teferi and showed that magmatism is dominated by Miocene aged basalts. In particular, Nelson et al. (2017) convincingly show that basalts from the Bale Mountains and Debre Sahil regions are less than 26 Ma old. Thus,

while earlier works (Zanettin et al., 1983) define the trap series as dominantly Oligocene–Miocene in age, the recent geochronology shows a distinct temporal break between the late Oligocene event and an emerging Miocene pulse of activity. Consequently, I will discuss the younger magmatism on the Somali Plateau in a subsequent chapter.

3.4. Flood basalts in Yemen

The 80,000 km^2 flood basalt occurrence in Yemen is generally thought to be contiguous with the northwestern Ethiopian Plateau. Thick, flat-lying basaltic trap series basalts capped with rhyolites are observed in Yemen to be ~1500–2000 m thick, though in places this thickness might exceed 3000 m, and may thin to just 300 m in the north of the province (Baker et al., 1996b; Chazot and Bertrand, 1993; Chiesa et al., 1989). The basalt portion of the sequence, which thickens towards the west and south of the province, lies upon a 200–400 m thick dominantly fluviatile/shallow marine sequence termed the 'Tawilah Group' (Baker et al., 1996a). The oldest flows from Yemen, which show some thickness variations suggestive of pre-existing topography (Baker et al., 1996b), are dated with ^{40}Ar–^{39}Ar techniques at 30.9 Ma (Baker et al., 1996a; Ukstins et al., 2002) similar to the basal sequences from the Ethiopian Traps (Hofmann et al., 1997). The basal flows appear to young towards the northeast of the province, where the lowest basalts record an ^{40}Ar–^{39}Ar date of 29.2 Ma (Baker et al., 1996a), consistent with a thin basal basalt layer and more voluminous rhyolites (Baker et al., 1996b). The earliest rhyolite activity at the top of the main phase basalts dates to 29.7 Ma (Riisager et al., 2005), thus bracketing the main basalt phase in Yemen to an interval equivalent to the Ethiopian Plateau, with the acknowledgement that the Ethiopian Traps may have commenced earlier (Ukstins et al., 2002).

Flood basalts from Yemen have been classified on the basis of the LT–HT scheme from the Ethiopian Plateau (Pik et al., 1998), into the HT1 and HT2 categories (Beccaluva et al., 2009; Natali et al., 2016). Similar to the northwest Ethiopian Plateau the flood basalts are typically evolved (less than 8% MgO) and have been impacted by fractional crystallization (Baker et al., 1996b; Chiesa et al., 1989). Some basalts have accumulated olivine, but as with the HT2 lavas from Ethiopia, near primary compositions are also observed (Baker et al., 1996b; Beccaluva et al., 2009; Kent et al., 2002; Natali et al., 2016).

Radiogenic isotope values encompass those from the northwest Ethiopian Plateau but extend to compositions that might suggest a more significant degree of crustal assimilation (^{87}Sr/^{86}Sr 0.7035–7055; ^{143}Nd/^{144}Nd 0.5129–0.5125; and ^{206}Pb/^{204}Pb 17.9–19.3) (Baker et al., 1996b; Baker et al., 2000; Kent et al., 2002). This concept of crustal assimilation was initially thought of as an upper crustal assimilant of Pan-African origin (Chiesa et al., 1989), but was later discussed by Baker et al. (1996b) in terms of contamination of the lavas by both the upper and lower crust. This AFC process is particularly evident in coupled variations of elevated δ^{18}O with increased ^{87}Sr/^{86}Sr and lower ^{143}Nd/^{144}Nd (Baker et al., 2000). Baker et al. (2000) identified three contaminating components: C1 — Archean Gneisses with distinctly unradiogenic Nd, and radiogenic Sr isotopic values; C2 — Pan-African lower crust which has an isotopic signature close to the basalt itself and is thus difficult to detect; C3 — Pan-African upper crust which has a radiogenic Pb isotope signature. Samples deemed by Baker et al. (2000) to be least contaminated by this process were similar to the Afar plume signature defined by Pik et al. (1999) for basalts on the northwest Ethiopian Plateau. The more 'contaminated' signature for the Yemen lavas might derive from the presence of Archean gneiss (C1 above), a lithology largely absent from the conjugate margin in Ethiopia. Kent et al. (2002) further probed the heterogeneity in the flood basalt lavas and found that primitive phenocrysts retained a wide variety of melt compositions while more evolved crystals contained more homogenous melt inclusions suggesting a melt aggregation process in the crust. Though these observations were described in terms of assimilation in the deep crust, 'exotic' Sr enriched inclusions hinted at heterogeneities in the lithology of the Afar plume (Kent et al., 2002).

Silicic volcanism in Yemen occupies a similar temporal window to that of the Ethiopian Plateau. Specifically, the appearance of rhyolite at the termination of the main phase basalts ca. 29.7 Ma (Riisager et al., 2005) and continuing to ca. 26.5 Ma (Baker et al., 1996a; Riisager et al., 2005; Zumbo et al., 1995), is parallel to the Oligocene ignimbrites from the Ethiopian Plateau (Ayalew et al., 2002; Sanders et al., 2004; Ukstins et al., 2002). Ukstins-Peate et al. (2005) have further refined these dates to suggest a pre-29.7 Ma Main Basalt phase, a 29.7–29.5 Ma 'Main Silicic' phase and a 29.5–27.7 Ma 'Bimodal' phase. Towards the terminal stages of the Main Basalt phase in Yemen, megacrystic plagioclase-glomerophyric basalt layers are evident, and are used to correlate the top of the flood basalt sequence throughout this area (Ukstins-Peate et al., 2005). West of Sana, and prior to the commencement of Main Silicic phase, a spatially restricted andesite has also been noted (Ukstins-Peate et al., 2005). During the Main Silicic phase, six pyroclastic flows are recorded that range in composition from trachyte to rhyolite and have been described in detail by Ukstins-Peate et al. (2005). Ukstins-Peate describes these eruptions as significant in volume and correlatable to those in Ethiopia, and even distally to sediments in the Indian Ocean. The subsequent Bimodal phase resulted in more volumetrically limited eruptions that were part pyroclastic, part effusive, and it is postulated that this transition is coincident with the commencement of rifting and extension in the region (Ukstins-Peate et al., 2005). Similar to the silicic rocks from the northwest Ethiopian Plateau, the origin of the rhyolite magmas is postulated to be either through fractional crystallization of the flood basalt magmas or melting of the underplate on the basis of oxygen isotopes (Baker et al., 2000). These rhyolites, like the underlying flood basalts, are prone to crustal assimilation and exhibit large ^{87}Sr/^{86}Sr–^{143}Nd/^{144}Nd excursions towards crustal components (Fig. 12). In other geochemical characteristics, the Yemen rhyolites overlap the Wengel Tena rhyolites from Ethiopia, consistent with a model for origin from the HT flood basalts.

3.5. Origin of the Oligocene flood basalts

The flood basalt episode reveals elevated mantle potential temperatures (T_P) in the source of these basalts. Specifically, values that range from 1440 to 1500 °C are recorded in this region, significantly elevated over ambient mantle values (Rooney et al., 2012b). In the adjacent Oligocene volcanism in Yemen, a parallel elevation in mantle potential temperature is evident with T_P in the range of 1500–1520 °C (Rooney et al., 2012b). These data indicate that the Oligocene flood basalt episode was coincident with a significant thermal anomaly within the mantle beneath the plateau. This interpretation is consistent with the large quantities of basalt erupted in a compressed time period (ca. 2 Ma: Hofmann et al., 1997). However, the mechanism by which

Fig. 12. Geochemical variation of silicic lavas during the Oligocene Traps phase and Early Miocene Resurgence phase in East Africa/Yemen. Isotopic values are those corrected for eruption age, where reported. Where uncorrected values for Pb and Nd isotopes are presented without an age correction, those values are used. However, we do not plot ^{87}Sr/^{86}Sr for samples where no correction has been applied given the significant ingrowth of ^{87}Sr that is possible in rhyolites since eruption. The isotopic reservoirs plotted: PA lithosphere — Pan-African Lithosphere; Afar Plume; and DMM — Depleted MORB mantle are from Rooney et al. (2012a). HIMU and EM2 are that of Hart et al. (1992). Data sources: Ali Adde — Deniel et al. (1994): Serae — Zanettin et al. (1999): Choke/Guguftu (note that sample 274 within the dataset from these authors is excluded as its strong positive Eu anomaly seem to point towards a cumulate origin) — Kieffer et al. (2004): Lake Tana — Dercq et al. (2001): Jimma — Ayalew et al. (2002): Wengel Tena — Ayalew et al. (2002); Ayalew and Yirgu (2003); Natali et al. (2011): Lima Limo — Ayalew et al. (2002); Ayalew and Yirgu (2003): Yemen — Ukstins-Peate et al. (2005); Baker et al. (2000); Chazot and Bertrand (1993, 1995); Manetti et al. (1991); Chiesa et al. (1989); Capaldi et al. (1983). Note that Choke/Guguftu, Serae, and Ali Adde are considered part of the Early Miocene Resurgence Phase while the rest are regarded as part of the tail end of the Oligocene Traps phase. The Lake Tana samples are of unclear temporal affinity but seem to plot with the Miocene samples.

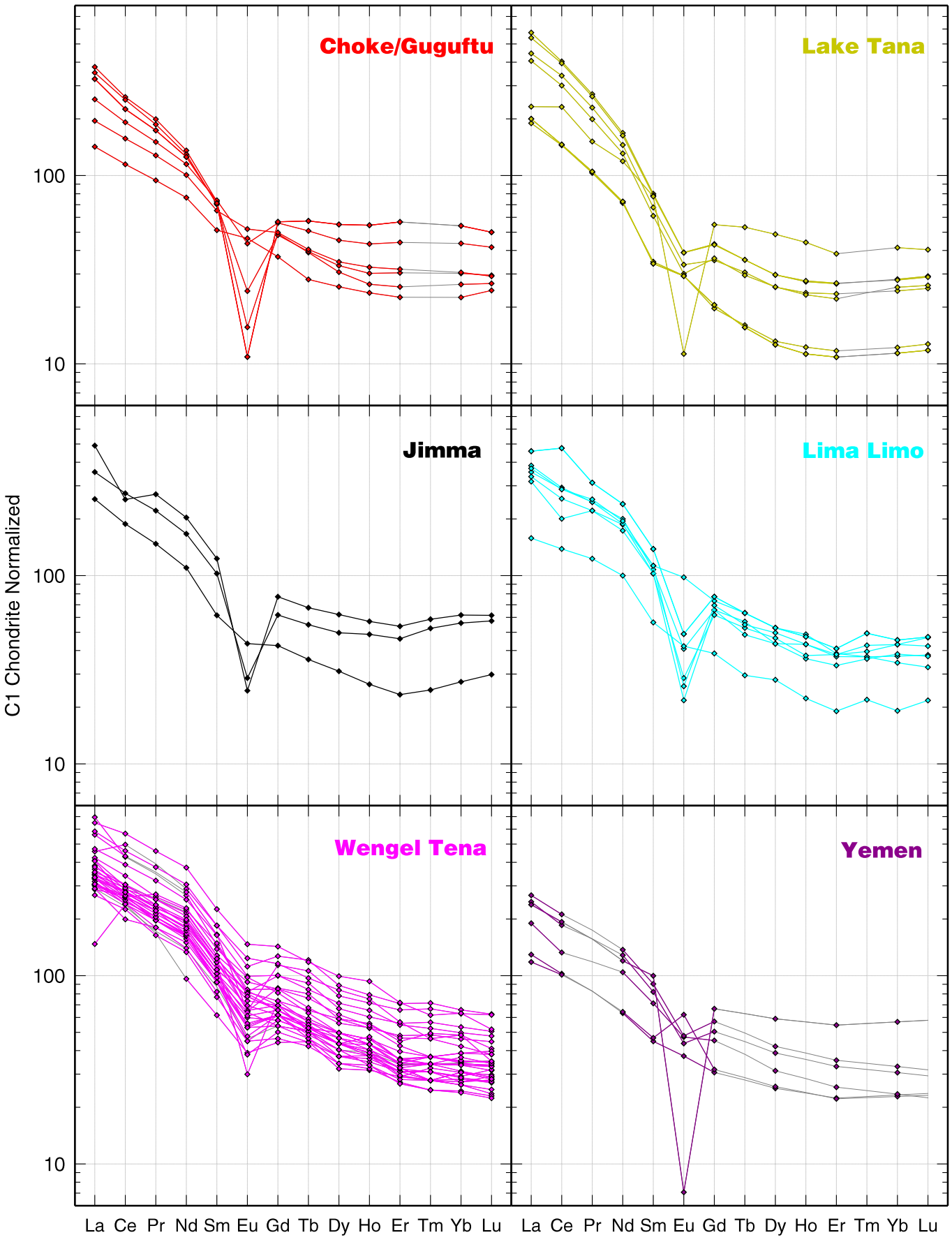


Fig. 13. Chondrite normalized data (Boynton, 1984) of silicic lavas erupted during the Oligocene Traps phase and Miocene Resurgence Phase in East Africa/Yemen. Data sources are the same as Fig. 12.

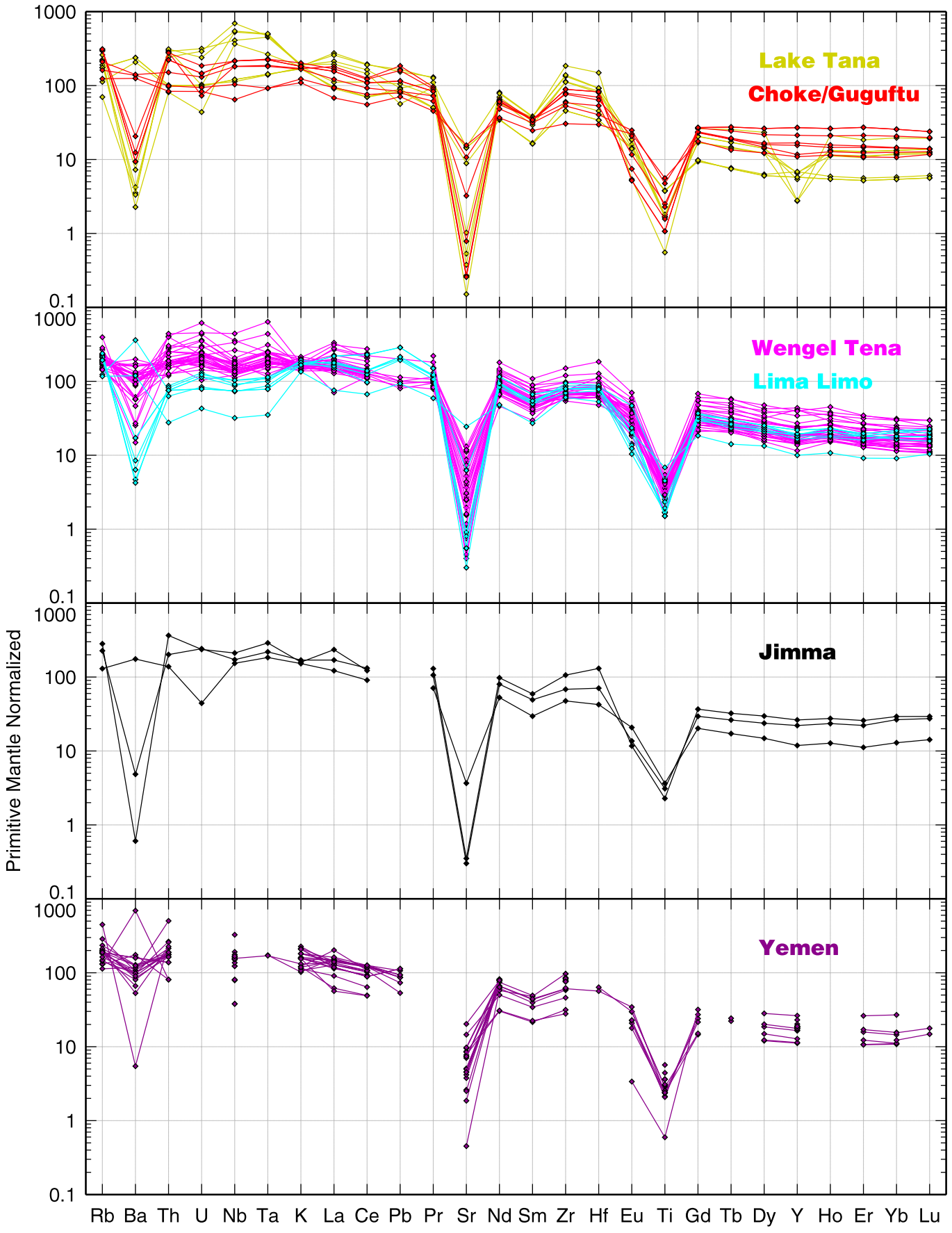


Fig. 14. Primitive mantle normalized data (Rooney et al., 2017) silicic lavas erupted during the Oligocene Traps phase and Miocene Resurgence Phase in East Africa/Yemen. Data sources are the same as Fig. 12. The unusual Y values for Lake Tana samples require further assessment as they differ from other lavas in the region.

magmas were generated remains controversial and is discussed in subsequent sections.

3.6. Post-flood basalt silicic and bimodal period: 29–26 Ma

Following the massive outpouring of basalts and late-stage silicic rocks in Ethiopia, Eritrea, and Yemen from ca. 31 to 29 Ma, there commenced a period of relative quiescence in these regions. As described previously, the termination of the flood basalt episode was accompanied by wide-scale silicic volcanism which is detected as far as the Indian Ocean in sediments (Ukstins et al., 2002), however this period was relatively short-lived and terminated by 29 Ma (Ukstins-Peate et al., 2005). During the following 2.5 Ma, more localized silicic eruptions accompanied by small basaltic flows, and diking along the nascent rift margin in Afar (Dessie) was the predominant magmatic activity throughout this region (Abbate et al., 2014; Baker et al., 1996b; Coulie et al., 2003; Rooney et al., 2013; Ukstins-Peate et al., 2005), though the aerial extent is small and not mapped.

Similar to the flood basalt activity in the North, the Turkana basin also exhibits a period of significant early Oligocene basaltic activity which is capped with silicic volcanic rocks, as I have previously described. The Mogila and Songot Hills region on the western edge of the Lotikipi plain best preserves the Oligocene sequence with 1800 m of basalt capped with 900 m of rhyolite. These rhyolites are temporally constrained with ^{40}Ar–^{39}Ar methods to have erupted between ca. 32.3–31.67 Ma (Brown and Jicha, 2016). In contrast, on the eastern side of the Lotikipi plain (Pelekech, Lokwanamur, and Moruerith Hills), similar thicknesses of basalts are overlain by tuffs and other silicic deposits of up to ~2 km in thickness, but are distinctly younger yielding ^{40}Ar–^{39}Ar ages between 25.9 and 27.2 Ma (Brown and Jicha, 2016). It is unclear as to whether older Oligocene rhyolites might be preserved within these thick rhyolite horizons. Rare instances of basaltic activity are evident during this period, and in Lothidok (closer to Lake Turkana) the Kalokol basalts underlie and overlie the well-studied fossiliferous Erageleit beds (Boschetto et al., 1992) and is a 785 m sequence of 20 olivine-augite basaltic flows and intercalated sediments (Fig. 9). Below the Eragaleit beds, the Kalokol basalts are only 100 m thick and record K–Ar dates of 26.9–28.0 Ma; the temporal break represented by the Eragaleit beds must be ca. 3 Ma, as the ~600 m thickness of basalts above these beds is ~24.1 Ma (Boschetto et al., 1992). It is therefore apparent that during this period, magmatic activity from Turkana to Yemen displayed a remarkably consistent tendency towards massive silicic eruptions that followed an Oligocene basaltic episode. Silicic volcanism then dominated magmatic activity with rare instances of basaltic volcanism until ca. 24 Ma. Differences in the age of the commencement of the large-scale silicic activity are apparent and young from south to north (ca. 32.3 Ma in Turkana, 30.2 Ma in Ethiopia and, 29.7 Ma in Yemen). However, this hiatus in basaltic magmatic activity was abruptly terminated ca. 24 Ma throughout the region.

4. Early Miocene resurgence phase

Throughout East Africa, the period following what has been considered the flood basalt episode was dominated by silicic eruptions with rarer intercalated basalts, with the exception of the significant basaltic episode evident in the Bale Mountains ca. 26 Ma. This status quo was disturbed ca. 24 Ma with a renewed period of basaltic activity that was broadly synchronous throughout the region (Abbate et al., 2014; Furman et al., 2006b; Kieffer et al., 2004). This period of volcanism is well-dated due to the tendency of these flows to cap important fossiliferous Miocene horizons — in particular in Eritrea (Abbate et al., 2014), and Turkana (Boschetto et al., 1992). Indeed, this Miocene activity is typically easily distinguishable from the underlying Oligocene flood basalts and silicic eruptions as flows over paleosols or angular unconformities. While there is some temporal variability in the commencement of these flows, I use the commonality of a significant basaltic eruptive event following a period of quiescence or silicic activity to define this event throughout the region (Fig. 15). It is notable that during this period, the first tentative evidence of magmatism south of Turkana is also recorded. Tuffs of carbonatitic affinity found within the Rungwe region (Fig. 2) have been dated using SIMS and $^{40}Ar/^{39}Ar$ techniques from 25.9–24.8 Ma (Roberts et al., 2012). Further work on this region is necessary to establish the temporal relationship of early carbonatitic magmatism in this region to the ongoing activity to the north.

4.1. Northwest Ethiopian Plateau

The most obvious manifestation of the ca. 24 Ma event on the Ethiopian Plateau are the imposing shield volcanoes of Choke (Fig. 15), which rises to over 4000 m in elevation and has a basal diameter of 100 km, and Guguftu which is rises to ~3900 m (Kieffer et al., 2004). Approximately 1000–1300 m of volcanic material has been built up within these shield volcanoes (Kieffer et al., 2004). Similar to the Oligocene Simien shield, the flows which make up Choke and Guguftu constitute thinner and less continuous magmatic activity, but importantly are more alkaline than the underlying flood basalts (Kieffer et al., 2004). Towards the base of Choke, a basaltic sample yielded a ~23 Ma date using ^{40}Ar–^{39}Ar techniques, but the Ar release plateau is not flat (Kieffer et al., 2004). The upper limit of magmatism at Choke and Guguftu are similar and are constrained by ^{40}Ar–^{39}Ar dates on rhyolites of 22.4 Ma and 23.3 Ma respectively (Kieffer et al., 2004). Kieffer et al. (2004) noted that Guguftu was unusual in having a significant amount of diking associated with the volcano, and some of these dikes from along the Afar margin have provided ^{40}Ar–^{39}Ar dates of 23.57 to 24.76 Ma (Rooney et al., 2013). Magmatic activity during this period was not restricted to the shield volcanoes and is also evident as ~200–400 m of flows over paleotopography to the west of Lake Tana dated using ^{40}Ar–^{39}Ar at 23.75 Ma (Prave et al., 2016). In the southern portions of the northwestern Ethiopian Plateau, a small eroded volcano (20 km diameter base) near Gerba Guracha with highly porphyritic flows and an intercalated lahar (similar to Choke) has been dated using ^{40}Ar–^{39}Ar methods between 24 and 25 Ma (Rooney et al., 2017). It is therefore apparent that throughout the northwest Ethiopian Plateau, a renewed period of widespread basaltic volcanism replaced the post-flood basalt silicic magmatic activity.

The petrographic and geochemical characteristics of these early Miocene shield basalts differ from most of the flood basalts underlying them in being porphyritic (dominantly plagioclase and olivine), alkaline, enriched in incompatible trace elements, and having a wide range of compositions extending from basalt to rhyolite (Kieffer et al., 2004; Rooney et al., 2013) (Figs. 12, 16). On primitive mantle normalized diagrams, basalts from Choke, Guguftu, and subset of those from Gerba Guracha have a distinctive positive anomaly in Ba and a depletion in U and Th that are similar to Oligocene HT1 basalts (Fig. 18) (Kieffer et al., 2004; Rooney et al., 2017). The Pb isotopic values in more primitive basalts from this group is much more radiogenic than the Oligocene HT1 basalts and ranges in $^{206}Pb/^{204}Pb$ from 19.0 to 19.6 (Fig. 16), while Sr and Nd isotopes do not greatly vary from HT1 basalts ($^{143}Nd/^{144}Nd$ = 0.51269–0.51295; $^{87}Sr/^{86}Sr$ = 0.7034–0.7044) (Kieffer et al., 2004; Rooney et al., 2013). Evolved magma compositions from this magma group exhibit significant isotopic variance from the more primitive magma compositions and are reflective of significant crustal assimilation impacting AFC processes (e.g., sample EAD-111: MgO 3.5 wt.%) which has $^{206}Pb/^{204}Pb$ = 18.2; $^{87}Sr/^{86}Sr$ = 0.7067; $^{143}Nd/^{144}Nd$ = 0.51259 (Rooney et al., 2013).

Silicic magmas from Choke and Guguftu are typically metaluminous to peraluminous and follow an evolutionary path in TiO_2 and Sr that resembles the Lima Limo (LT) rhyolites from the Oligocene Traps phase (Fig. 12). However, silicic magmas during the Miocene resurgence phase differ in rare earth element patterns from the Oligocene Traps phase rhyolites (Fig. 13). In particular, the light and middle rare earth elements (e.g. La/Sm; Fig. 12) are significantly more fractionated

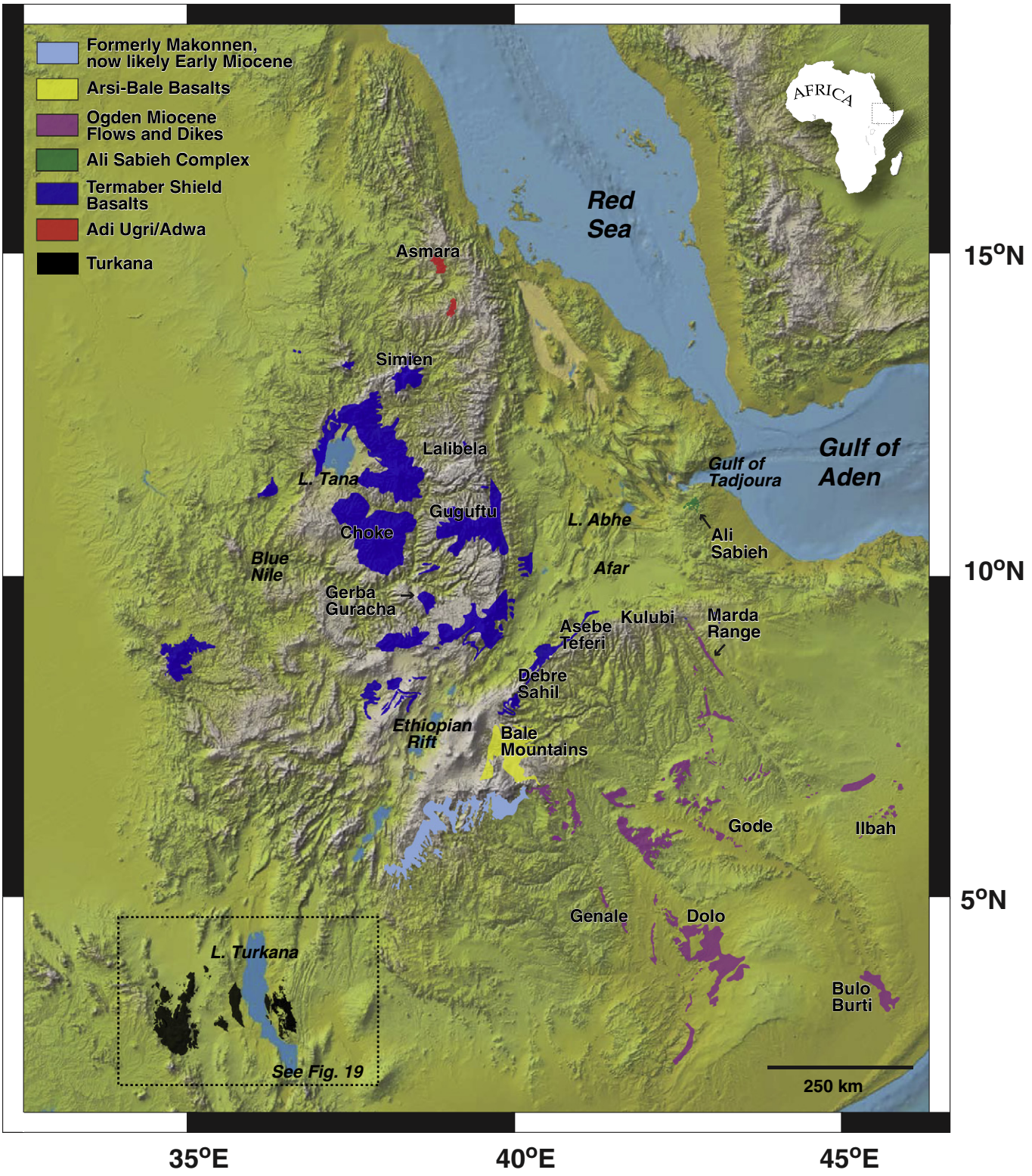


Fig. 15. Magmatic activity in East Africa/Yemen during the Early Miocene Resurgence Phase (26.9–22 Ma). Data sources the same as in Fig. 3. Place names from Ogaden from Mège et al. (2015). The inset box refers to Fig. 19.

during the Miocene resurgence phase. While the Lake Tana rhyolites (Dercq et al., 2001) are of unknown temporal affinity, their patterns (Figs. 12, 13, 14) might suggest linkage with the Miocene events but more work is needed in that region.

A second group of primitive magmas has been identified from the Gerba Guracha volcano that resemble HT2 flood basalts in that they have extreme enrichment in TiO_2 (up to 6.5 wt.%; Fig. 16) and in many incompatible trace elements (Fig. 17), and exhibit a pronounced negative K anomaly on primitive mantle normalized diagrams (Fig. 18: Rooney et al., 2014a, 2014b; Fig. 17). These lavas do, however, differ from the Oligocene HT2 basalts in that the high-Ti Gerba Guracha lavas are extremely clinopyroxene (and Fe–Ti oxide) phyric but

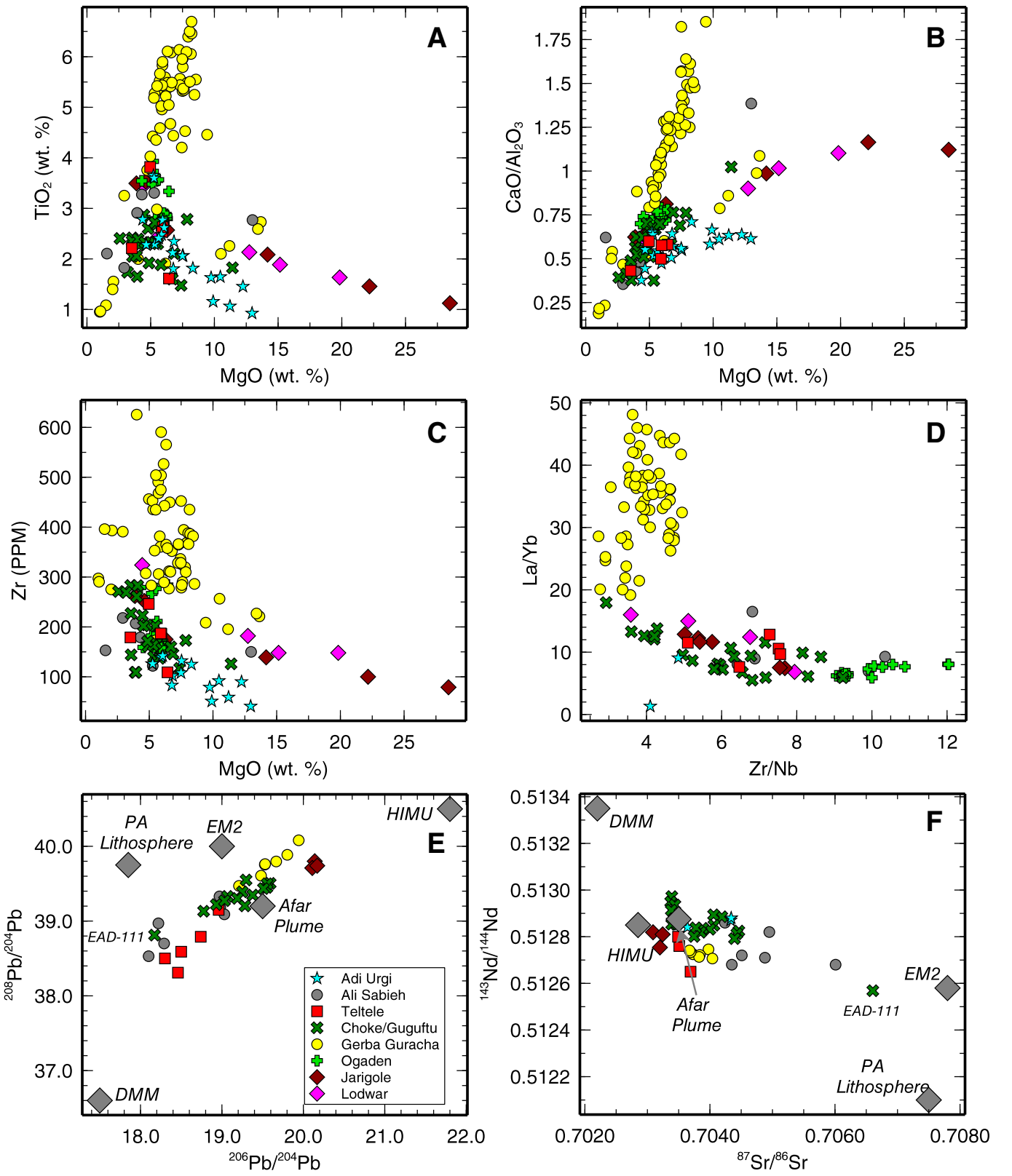


Fig. 16. Geochemical variation of mafic lavas during the Early Miocene resurgence phase in East Africa/Yemen. Isotopic values are those corrected for eruption age, where reported. The isotopic reservoirs plotted (PA lithosphere — Pan-African Lithosphere; Afar Plume; and DMM — Depleted MORB mantle is from Rooney et al. (2012a). HIMU and EM2 are that of Hart et al. (1992). Data sources: Adi Urgi (Eritrea) — Teklay et al. (2005); Zanettin et al. (1999): Choke/Guguftu Shield Volcanoes (Ethiopia) — Kieffer et al. (2004); Rooney et al. (2013): Gerba Guracha (Ethiopia) — Rooney et al. (2014b, 2017): Ogaden — Mège et al. (2016): Jarigole and Lodwar — Furman et al. (2006b): Teltele — Stewart and Rogers (1996): Ali Sabieh — Deniel et al. (1994). Note samples EAD-111 is discussed in the text and is from Rooney et al. (2013). Nelson et al. (2017) is currently in review, thus data from that manuscript is not included in these plots. Information from that manuscript is used with the permission of Wendy Nelson.

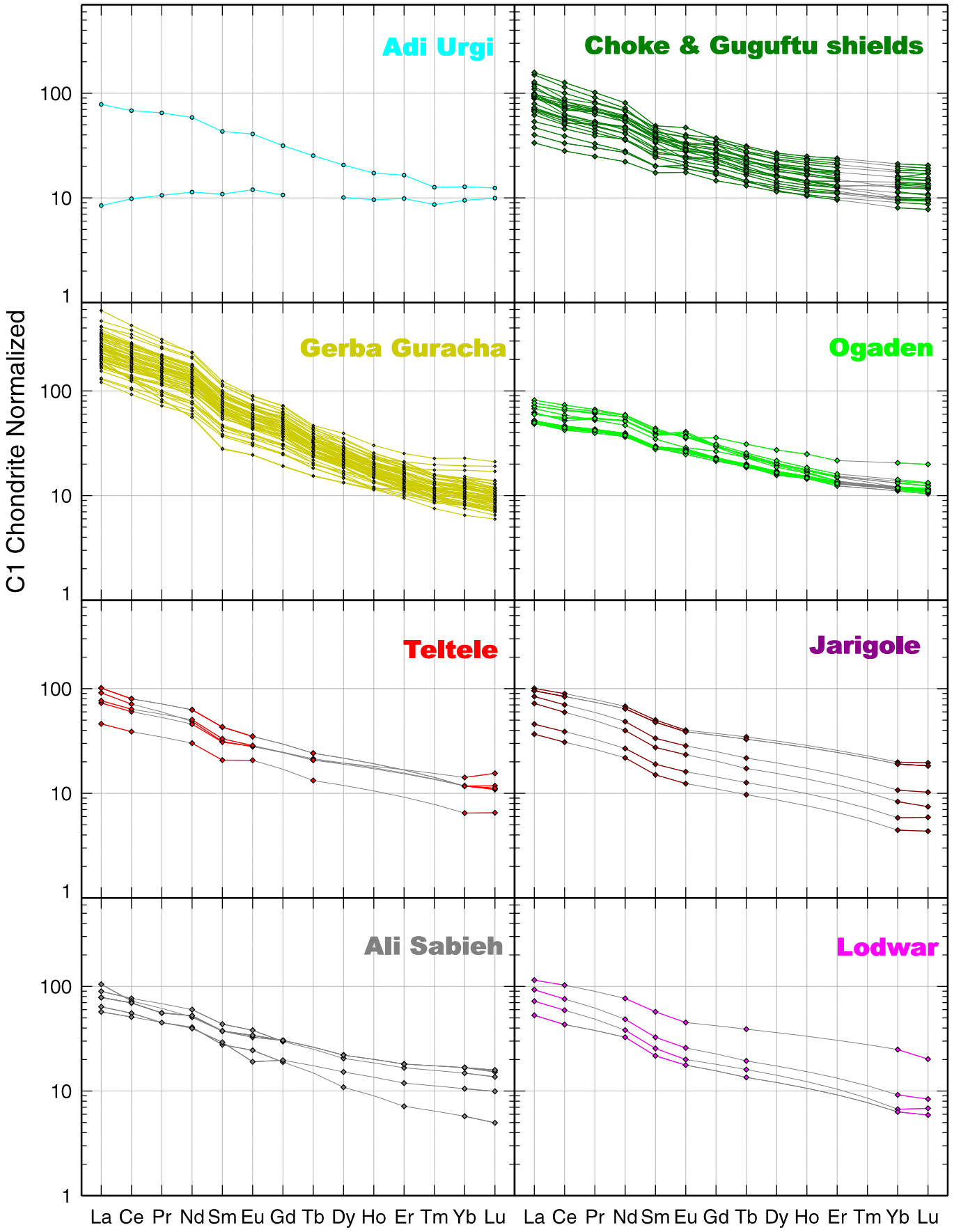


Fig. 17. Chondrite normalized data (Boynton, 1984) of mafic lavas erupted during the Early Miocene resurgence phase in East Africa/Yemen. Data sources are the same as Fig. 16. Note that the value of Tb is not included in one of the Adi Urgi samples as it is anomalous and was not plotted in the original publication (Teklay et al., 2005).

typically lack olivine (in contrast to the olivine-rich textures of the HT2 lavas) (Rooney et al., 2017). More distinctly, the Gerba Guracha high-Ti lavas are profoundly silica undersaturated (as low at 35.4 wt.% SiO_2), have elevated CaO (up to 17.1 wt.%), and have a consistently high loss on ignition (~3–5%) despite what has been petrographically-defined as being largely unaltered rock. These characteristics are suggestive for the influence of carbonate in the origin of the high-Ti Gerba Guracha lavas (Rooney et al., 2017). Some compatible trace elements are also extremely depleted in these rocks with less than 100 ppm Ni and Cr despite a range in MgO extending to ~9 wt.%. These rocks are also unusual isotopically, and have significantly more radiogenic $^{206}Pb/^{204}Pb$ of up to 20 in comparison to the lower Ti variety of early Miocene basalts noted above (Fig. 16). These lavas plot below the ε_{Hf}–ε_{Nd} mantle array, and are thus indicative of an origin with a HIMU-like source (Rooney et al., 2014b). Rooney et al. (2014b, 2017) showed that the isotopic and geochemical signatures evident in the high-Ti Gerba Guracha group are consistent with evolution within an amphibole-rich lithospheric mantle metasome generated contemporaneously with the construction of the regional lithosphere during the Pan-African orogenic events, raising the possibility that there is a HIMU-like reservoir in the East African lithospheric mantle.

At the northern edge of the northwest Ethiopian Plateau, early Miocene magmatism is represented by the 26.94 (±0.36) Ma Adwa trachytes (Natali et al., 2013). These rocks, which are more alkaline than the preceding flood basalts, are considered to have been derived from a smaller degree of melting in the mantle followed by significant fractional crystallization and eruption along pre-existing lineaments (Natali et al., 2013). Further to the north, in Eritrea, magmatism during this period is more mafic and is represented by the alkaline Adi Ugri basalt (locally known as Ono and Durko basalt: Teklay et al., 2005) which yields an ^{40}Ar–^{39}Ar age of 23.55 Ma at its base (Abbate et al., 2014) and ^{40}Ar–^{39}Ar & K–Ar ages of 20–22.72 Ma higher in the sequence (Teklay, 2003 cited in Teklay et al., 2005; Zanettin et al., 2006). The Adi Ugri basalts are intercalated with sediments and at least one tuff suggesting periods of quiescence in between the eruptive episodes (Zanettin et al., 1999). Zanettin et al. (2006) note that the Adi Ugri basalt has a fine grain size and more fissural eruptive characteristics in comparison to the shield volcano derived porphyritic lavas in Ethiopia. The lavas span a wide array of compositions (4.7–12.5 wt.% MgO; Fig. 16), are transitional in terms of alkalinity (Teklay et al., 2005; Zanettin et al., 1999), and are similar in composition to the contemporaneous Choke and Guguftu shield volcanoes in Ethiopia. However, there appears to be significant unresolved heterogeneity in the trace element data given the range of $(La/Yb)_N$ is 0.9 to 6.1 (Fig. 17). The isotopic constraints for these basalts are within the range of Choke and Guguftu and range from $^{87}Sr/^{86}Sr$ 0.7034–0.7043 and $^{143}Nd/^{144}Nd$ 0.51284–0.51288 (Fig. 16: Teklay et al., 2005). Rhyolites, which occur within the Adi Urgi sequence (Serae rhyolites: Zanettin et al., 1999, 2006) are unusual in that they are peralkaline (Fig. 12), the only rhyolites known to be of this composition during this period. The Serae rhyolites have low TiO_2, but the lack of trace elements or isotopic information prevents further classification. Given the evidence of trace element heterogeneity beyond that observed in the contemporaneous Choke and Guguftu shields, more investigation of the Adi Ugri basalts and their associated rhyolites is necessary to ascertain a link between this activity and that further south.

4.2. Yemen

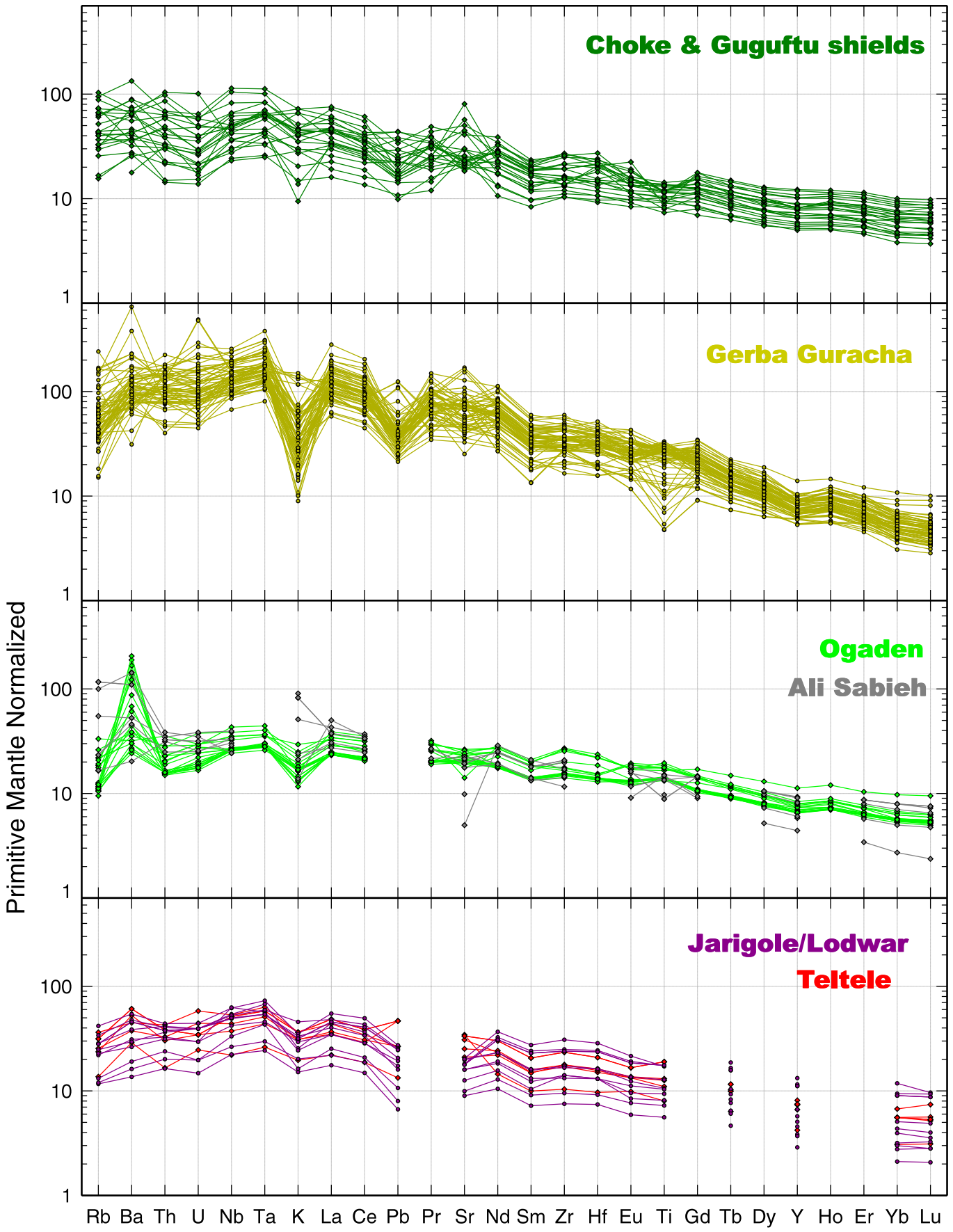
Curiously, there is a lack of widespread early Miocene basaltic activity in Yemen. While there are some instances of isolated occurrences of potentially Miocene basalt (Mattash et al., 2014), they have not been stratigraphically separated from the underlying flood basalts, accordingly the wide spread of K–Ar ages obtained (35–16 Ma) make it difficult to assess any correlation with northwest Ethiopia. Whereas stratigraphic and detailed ^{40}Ar–^{39}Ar studies have been undertaken on continuous sections, the relative paucity of volcanic rocks in Yemen from 26.5 Ma to 19.7 Ma (synchronous with the breakup unconformity: Ukstins et al., 2002) is thought to relate to the commencement of rifting ca. 26 Ma (Ukstins et al., 2002; Ukstins-Peate et al., 2005). However, magmatism during this magmatically quiet period is present on the Arabian plate along a long (1700 km), but narrow stretch of the Red Sea Coast. ^{40}Ar–^{39}Ar data derived from rocks in this region have revealed a clustering of ages from 24 to 21 Ma (Féraud et al., 1991; Sebai et al., 1991). While these authors suggest a linkage between this magmatism and the opening of the Red Sea, it is notable that this activity along the Red Sea margin is synchronous with the resurgence in basaltic volcanism identified throughout north-east Africa at this time. An evaluation of these Red Sea margin rocks, which extend into Saudi Arabia and further north, is beyond the scope of this synthesis.

4.3. Somali Plateau

Magmatic activity on the Somali Plateau is distinct from that on the northwest Ethiopian Plateau in that the earliest phase of volcanism in this region appears to be dominantly Early Miocene in age, with the exception of some late Oligocene HT1 flows at Asebe Teferi, near the Afar margin (Nelson, 2009; Nelson et al., 2017). While there has been a long-standing understanding that the Somali Plateau activity post-dated eruptions on the northwest Ethiopian Plateau (Abbate and Sagri, 1980; Jones, 1976), it is only with recent ^{40}Ar–^{39}Ar dates at the base of the basalt pile that these basalts could be properly correlated. Jones (1976), using K–Ar techniques suggested that the earliest stages of volcanism on the Somali Plateau were ca. 23–24 Ma. This has been further refined by Chernet et al. (1998) who using K–Ar dating established a period of volcanism ca. 24 Ma along the margin of Afar near Kulubi. This confirms the mapping of Juch (1978) who assigned flows in this area to the UTB (see Section 3.3), and distinctly different from initial studies in the region who suggested a Mesozoic age on the basis of stratigraphic constraints (Canuti et al., 1972). Importantly, however, the work by Canuti et al. (1972) shows that the 60 m thick Early Miocene flows and volcanic tuffs/breccias in this region erupted directly on the Mesozoic Antalo limestone, and were capped later by a sandstone, further demonstrating the limited volume of Oligocene material on the Somali Plateau. Nelson et al. (2017) found using ^{40}Ar–^{39}Ar techniques that activity is ca. 24.5 Ma at Debre Sahil, adjacent to the Central Main Ethiopian Rift, and 25–26 Ma in the Bale Mountains, which is towards the eastern edge of the Somali Plateau. Thus, basaltic activity on the Somali Plateau extends to 26 Ma, making it unusual and among the few known regions with basaltic volcanism dated to this period. I have grouped this magmatism in the Bale Mountains with the renewed ~24 Ma basaltic event given that basaltic volcanism is continuous from 26 Ma through the renewed activity at 24 Ma seen elsewhere.

Unlike the shield volcanoes that developed on the northwestern Ethiopian Plateau during this period, magmatic activity on the Somali Plateau was fissural (Abbate and Sagri, 1980). At volcanic centers up to 2000 m of volcanic material may accumulate but typically flow thicknesses are ca. 200–500 m (Mège et al., 2015). Despite a significant age difference, the basalt unit at the Bale Mountains (ca. 26 Ma) is similar to the older 28.6 Ma Asebe Teferi unit (i.e. aphyric); these basal layers are overlain by felsic units and porphyritic basalts (Nelson, 2009; Nelson et al., 2017). The 26–24 Ma lavas from Debre Sahil (Fig. 15) and the Bale Mountains (Fig. 15) are alkaline and exhibit a

Fig. 18. Primitive mantle normalized data (McDonough and Sun, 1995) of mafic lavas erupted during the Early Miocene resurgence phase in East Africa/Yemen. Data sources are the same as Fig. 16. Note that lavas from Eritrea have been highlighted in cyan due to their distinction from other LT flood basalts. Note that Adi Urgi was not plotted due to a lack of data beyond the rare earth elements available for these samples.


Choke & Guguftu shields
Gerba Guracha
Ogaden
Ali Sabieh
Jarigole/Lodwar
Teltele
Primitive Mantle Normalized
100
10
1
Rb Ba Th U Nb Ta K La Ce Pb Pr Sr Nd Sm Zr Hf Eu Ti Gd Tb Dy Y Ho Er Yb Lu

wide range of compositions extending from basalts/basanites to basaltic trachyandesites (Nelson, 2009; Nelson et al., 2017). Lavas from Debre Sahil exhibit similar trace element patterns to the contemporaneous Choke and Guguftu shield volcanoes (i.e. positive Ba, and negative Th–U anomalies on a primitive-mantle normalized diagram). Radiogenic isotopes for these rocks ($^{87}Sr/^{86}Sr$ 0.7039–0.7041; $^{143}Nd/^{144}Nd$ 0.5127–0.5128; $^{206}Pb/^{204}Pb$ 19.3–19.6) are also comparable with magmatism from the Choke/Guguftu shields.

By contrast, Nelson et al. (2017) showed that lavas from the Bale Mountains show a temporal evolution:

(A) lower flows exhibit significant enrichment in incompatible trace elements and a negative K anomaly in primitive mantle normalized diagrams similar to Miocene flows from Gerba Guracha. Isotopically, these flows also resemble the contemporaneous lavas from Gerba Guracha and display $^{206}Pb/^{204}Pb$ of 19.8–20.1, $^{87}Sr/^{86}Sr$ ~ 0.7033–0.7036; $^{143}Nd/^{144}Nd$ ~ 0.5127. Importantly, the ε_{Hf}–ε_{Nd} characteristics of these samples places them below the mantle array (distinct from any Oligocene basalt) and again parallels observations from Gerba Guracha and are consistent with a derivation within the lithospheric mantle.

(B) Younger flows exhibit a humped primitive mantle normalized diagram paralleling Miocene picrites from Turkana (e.g., Furman et al., 2006b). However, the isotopic characteristics are less extreme with $^{206}Pb/^{204}Pb$ 19.2–19.3; $^{87}Sr/^{86}Sr$ ~ 0.7035; $^{143}Nd/^{144}Nd$ ~ 0.5128, plotting broadly within the Afar plume endmember, similar to more primitive lavas from the Oligocene period.

In summary, magmatism on the Somali Plateau has some distinctions to that on the northwest Ethiopian Plateau in that Oligocene flood basalts do not constitute the most significant volume of the Somali Plateau, and there is a lack of widespread silicic activity following the termination of the flood basalts. However, the ca. 25–26 Ma resurgence in basaltic volcanism appears to be a common feature of both plateaus, potentially earlier on the Somali Plateau. Given the evidence of significant magmatic activity during the early Miocene along the Somali Plateau, it is therefore unsurprising that the distal eruptions within Ogaden/Somaliland (east of the Somali Plateau towards the Indian Ocean) also date to this early Miocene magmatic resurgence (e.g., Mège et al., 2015).

To the east of the Somali Plateau lies the lowland area of Ogaden where flows continue for a further 600 km (Fig. 15) but at a much reduced thickness not exceeding 50 m, frequently following rivers or creating table lands (Mège et al., 2015). These low viscosity flows extended 100–200 km and can be traced today as 'meandering hills' (Mège et al., 2015). In contrast, the uplifted Marda Range, which is the manifestation of a major shear zone (Purcell, 1976), is as a 150 km long, 2 km wide remnant of Mesozoic sediments capped by a 200 m flow (Mège et al., 2015). Mège et al. (2016) indicate that the dike swarm identified near the Ethiopia–Sudan border is contiguous with the Marda range and forms part of a larger 'Ogaden Dike Swarm'. ^{40}Ar–^{39}Ar dates for these dikes are 26.6–22.6 Ma (Mège et al., 2016), and are thus contemporaneous with magmatic activity on the adjacent Somali Plateau (Nelson, 2009; Nelson et al., 2017). The geochemical characteristics of these dikes are variable with no clear temporal progression (Mège et al., 2016). All classify roughly as HT1, though this division can also incorporate some magmas from the Miocene Choke and Guguftu shields. While overlap with the Choke and Guguftu shields is evident, the Ogaden basalts plot at slightly more depleted values for incompatible trace elements with high values of Zr/Nb (10–12), and more depleted profiles in chondrite normalized (Fig. 17) and primitive-mantle normalized diagrams (Fig. 18). Mège et al. (2016) suggest that the Ogaden dike swarm is part of a major extensional event during the early Miocene and may be related to the contemporaneous Red Sea dike swarm.

4.4. Turkana

In the Turkana basin, the 25–23 Ma period continued to record silicic volcanism in Pelekech and Morueith/Kalin; basaltic volcanism is apparent further to the east, but is spatially restricted to the shores of Lake Turkana at about its mid length in the Lothidok Range (western shore) and in the Jarigole Range (eastern shore) (Brown and McDougall, 2011). While Pelekech and Morueith/Kalin have been broadly assigned to the Oligocene period, the stratigraphic sequence records continued activity here to ca. 19 Ma (Morley et al., 1999). The Jarigole Volcanics are considered correlative with the Emuruabwin Tephrites (Fig. 19), which crop out on the western side of Lake Turkana (Ochieng' et al., 1988).

Within the Lothidok range, basaltic eruptions re-commenced after a ca. 3 Ma hiatus that is represented by the fossiliferous Eragaleit beds (Boschetto et al., 1992). These ~500 m thick upper Kalakol basalts commenced eruption at 24.2 Ma, and continued (with some sedimentary horizons) until ca. 17.9 Ma (Boschetto et al., 1992). On the opposing (eastern) shore of Lake Turkana, the Jarigole Range (extending to Alia Bay: Wilkinson, 1988) also contains an array of silica under-saturated compositions from phonolite to olivine nephelenites and tephrites (Brown and McDougall, 2011; Wilkinson, 1988). Morley et al. (1992) have dated these rocks to 23.2 Ma and the 400 m sequence of the Jarigole Volcanics is considered to extend to 17 Ma (Ochieng' et al., 1988).

The eruptions in Turkana during this period are thought to have derived from a mantle with an elevated mantle potential temperature (1513 °C: Rooney et al., 2012b). The geochemical characteristics of samples from the Lothidok Range (Lodawar) and Jarigole Range suggest similarities in the parental magmas for the lavas erupted contemporaneously in this region. The trace element pattern roughly parallels that of the earlier Balesa Koromto Type B lavas erupted 40 km further south during the Eocene (Furman et al., 2006b). However, the overall incompatible trace element enrichment of the Early Miocene samples (Figs. 17, 18) is significantly less than that of the Eocene eruptive phase, with a much less pronounced negative K anomaly (Furman et al., 2006b). The isotopic characteristics of the Early Miocene rocks (Fig. 16) parallel that of Balesa Koromto lavas in having particularly elevated $^{206}Pb/^{204}Pb$ (20.1–20.3) and low $^{87}Sr/^{86}Sr$ (0.7032–7033). The Jarigole samples plot below the ε_{Hf}–ε_{Nd} mantle array (see Section 5) and have low $^{3}He/^{4}He$ of 6.2–6.6 R/R_A (Furman et al., 2006b; Nelson et al., 2012), further supporting an origin of these rocks within a HIMU-like source similar to those noted in the Miocene of Ethiopia (Rooney et al., 2014b).

4.5. Southern Ethiopia

Following a long hiatus, magmatic activity once again commenced in Southern Ethiopia with the Teltele basalts, which erupted over crystalline basement or the overlying sediments (Davidson, 1983). These basalts, which erupted on the eastern side of the Chew Bahir basin on the Teltele horst (Fig. 15), form a 400 m thick sequence of flows, with a ^{40}Ar–^{39}Ar date from the basal flows of 23.0 Ma (Ebinger et al., 2000). This basalt has been amalgamated by George and Rogers (2002) with the Getra Kele basalt that occurs near the Chamo border fault on the western side of the Ethiopian Rift (Fig. 4). However, the Getra Kele Basalt has a slightly younger ^{40}Ar–^{39}Ar age of 19.1 Ma, but continuing through 12.4 Ma (George et al., 1998). The Getra Kele basalt, which is described as syn-extensional, thus falls outside the purview of this review. However, I have added the Teltele basalts, which plot with many of the other Miocene basalts. The existing geochemical data for the Teltele basalts points to similarities with the earlier Amaro and Gamo lavas — specifically, the pattern of Ba enrichment and broadly parallel primitive mantle normalized values for these basalts (Fig. 18). Isotopically, these samples are also similar with $^{87}Sr/^{86}Sr$ ~ 0.7035–0.7037, $^{143}Nd/^{144}Nd$ ~ 0.51265–0.51280, and $^{206}Pb/^{204}Pb$ 18.3–19.0

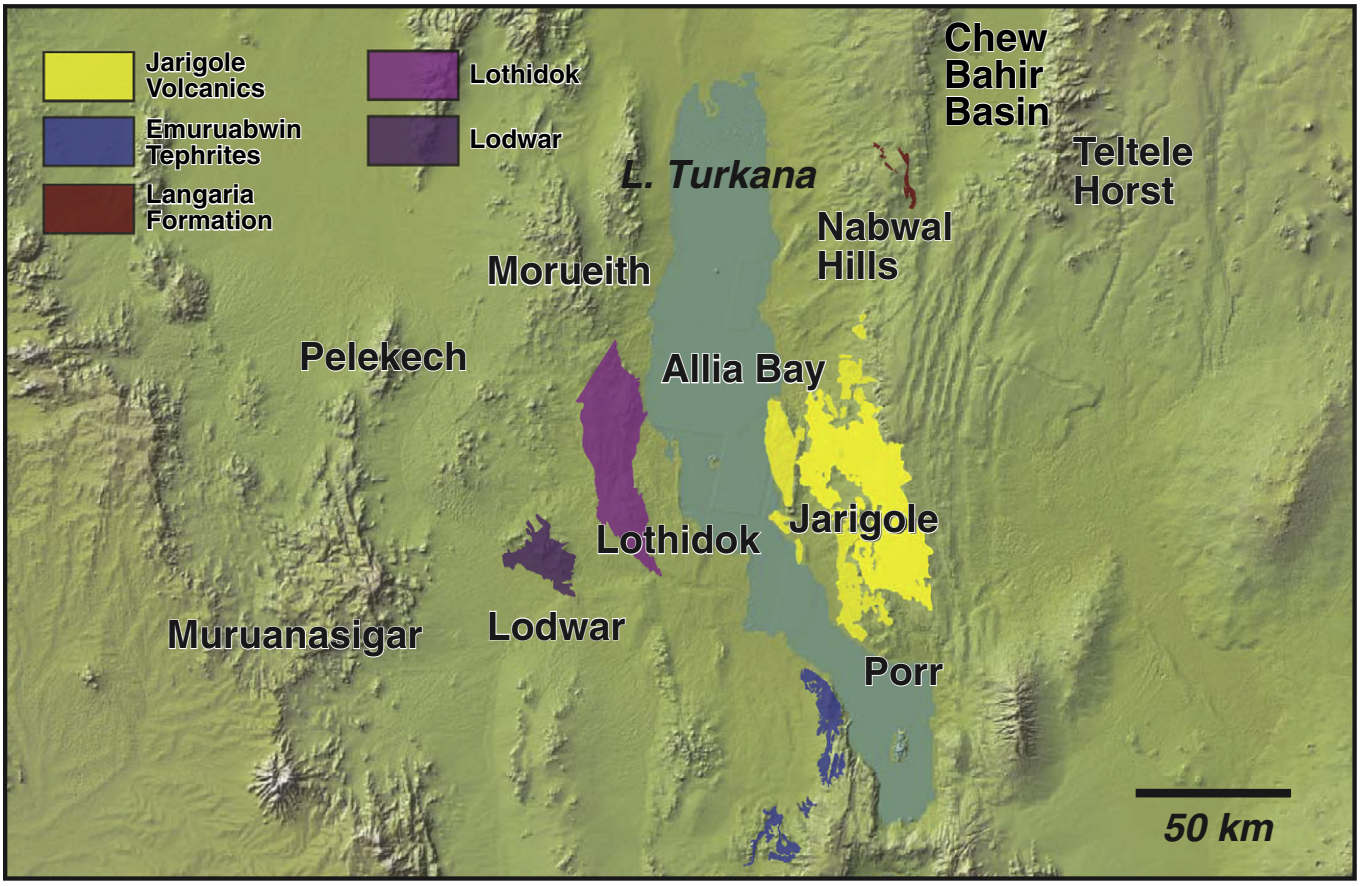


Fig. 19. Magmatic activity in the Turkana Basin during the Early Miocene Resurgence Phase (26.9–22 Ma). Data sources the same as in Fig. 3.

(Fig. 16). It is therefore apparent from a trace element and isotopic perspective that the Teltele basalts were generated within the same mantle source as the previous episodes of activity in Southern Ethiopia.

4.6. Afar — the beginning of the depression

For the first time, volcanism is recorded in the Afar Triangle during this time period. The compilation of K–Ar dates by Barberi et al. (1972) shows a significant pulse of activity ca. 22–25 Ma, much of it silicic in composition. Basalts intercalated with the 'Red Series' (the oldest detrital sediments found within Afar) provide a ca. 24 Ma date for volcanism in this region and has been used to suggest that the Danakil block had subsided by this time (Barberi et al., 1972). In Djibouti, the contemporaneous basaltic sequence is known as the Ali Sabieh or Adolei basalts or the 'Old Basalts', which are interbedded with the Ali Adde rhyolites (Deniel et al., 1994). Zumbo et al. (1995) obtained a ^{40}Ar–^{39}Ar date of 23.6 Ma for these basalts, consistent with the previous K–Ar dates from Afar broadly (Barberi et al., 1972). Further work by Le Gall et al. (2010) suggests that the Ali Sabieh basalts are actually intrusive bodies and provided further K–Ar dates of between 20 and 27 Ma for they termed 'the main mafic intrusion'. The Ali Sabieh basalts/main mafic intrusion are distinctive in being among the most crustally contaminated mafic rocks generated up to this time within the province, displaying variability in large ion lithophile elements (Fig. 18) and evidence of a significant crustal overprint on radiogenic isotopes (Fig. 16: $^{87}Sr/^{86}Sr$ 0.7042–0.7060; $^{143}Nd/^{144}Nd$ 0.51268–0.51286; $^{206}Pb/^{204}Pb$ 18.1–19.0), consistent with evidence for significant crustal contribution in contemporaneous Ali Adde rhyolites (Fig. 12: Deniel et al., 1994). A similar pattern of crustal assimilation coincident with episodes of rifting is evident in the later Burka formation (20–13 Ma), located along the eastern margin of the northwest Ethiopian Plateau (Rooney et al., 2013) but is not addressed in this review.

4.7. End of an era

The increase in extension at ca. 25–22 Ma due to slab pull from the Urumieh-Dokhtar arc forming as a result of the closure of the Neotethys, resulted in widespread marine deposition within the evolving Red Sea (Bosworth et al., 2005). Synchronous with this extensional event is the tilting of Oligocene flood basalts along the nascent Afar margin as a result of crustal stretching (Stab et al., 2016), and the formation of a rift-parallel dike swarm and eruptions along the margin of the Red Sea that are interpreted to be derived from de-stabilization of the extending lithosphere (Abu El-Rus and Rooney, 2017). This period marked the end of region-wide synchronous events. The initial flood basalt pulse in Southern Ethiopia and Turkana, followed by the voluminous Oligocene eruptions and terminated by the Early Miocene flows were distinctive due to the wide areas over which the events synchronously manifested. Subsequent magmatic activity within the rift became more localized and controlled by extensional structures. Consequently, I will address this period of extension-dominated magmatic activity in Part II of this synthesis.

5. Magma types in the East African rift

From 45 Ma to 22 Ma, lavas erupted within East Africa exhibit commonalities in terms of their major element, trace element, and isotopic characteristics. Indeed, a number of distinctive magma 'families' can be identified. The identification of these magmas facilitates probes of the origin of magmatism and is thus useful in understanding how melt has been generated in the province. Primitive mantle normalized

spider diagrams provide an opportunity to examine the trace element behavior and on the basis of these patterns there are three apparent families of magmas.

5.1. Group I — an incompatible trace element depleted family of magmas

5.1.1. Group Ia: the Oligocene LT group

The group Ia family of magmas are typified by the Early Oligocene LT flood basalts. These lavas are distinguished by their enrichment in Ba and Pb but with an otherwise flat trace element pattern roughly following a line denoting 10× enrichment over primitive mantle (Fig. 20). Overall, lavas within the group Ia family have the least enrichment in the incompatible trace elements. There is a moderate relative depletion in $(Tb/Yb)_{CN}$ of ~1.01–1.84 (median of 1.58). The group Ia magmas are temporally and spatially restricted in their occurrence. These lavas are found only during the Early Oligocene flood basalts event and only towards the western side of the northwestern Ethiopian Plateau and include the Simien Shield. One exception is the two mafic dikes from the Afar margin which are post-flood basalt and dated ~29.8–27.4 Ma (Rooney et al., 2013). Given that these magmas constructed a large portion of the northwest Ethiopian Plateau, the spatial–temporal restriction on their appearance does not impact their volumetric contribution to magmatism regionally.

Isotopically, the group Ia family of magmas are:

(i) Unradiogenic in terms of Pb isotopes: $^{206}Pb/^{204}Pb$ ~ 18–18.5; $^{207}Pb/^{204}Pb$ ~ 15.51–15.55; $^{206}Pb/^{204}Pb$ 37.6–38.6, but with exceptions in Sudan and along the margin of the Ethiopian Rift (Lucassen et al., 2008; Rooney et al., 2013)
(ii) Moderately radiogenic in terms of Sr isotopes: $^{87}Sr/^{86}Sr$ 0.7036–0.7045, but the majority less than 0.7040 (Fig. 10).
(iii) Moderately radiogenic $^{143}Nd/^{144}Nd$: 0.51280–0.51296
(iv) Low $^{3}He/^{4}He$ of ~4.4 R/R_A (Pik et al., 2006)

5.1.2. Group Ib: the LILE depleted group

The Amaro and later Gamo groups of Southern Ethiopia share similar trace element profile to the Oligocene LT flood basalts from the northwestern Ethiopian Plateau that comprise group Ia. In particular, these groups share a broad enrichment in Ba, however the lack of positive anomalies in K, Sr, and Pb (Fig. 20) are a notable distinguishing feature between groups Ia and Ib. Group Ib magmas have overall a flatter trace element pattern, less prone to enrichment in LILE in comparison to group Ia.

I have tentatively assigned Gamo lavas to group IIb given the geographic proximity to the Amaro lavas, but the Gamo lavas do differ in distinctive ways. Notably, the Gamo lavas follow the same broad pattern as the Amaro group but are significantly more enriched in the most incompatible trace elements. Thus, the Gamo group appears displaced to higher concentrations of these elements. The origin of the Amaro/Gamo lavas is considered to be related to a mantle plume in this region (George and Rogers, 2002).

Isotopically, the group Ib family of magmas are:

(i) Moderately unradiogenic in terms of Pb isotopes: $^{206}Pb/^{204}Pb$ ~ 18–19; $^{207}Pb/^{204}Pb$ ~ 15.50–15.61; $^{208}Pb/^{204}Pb$ 37.9–38.9
(ii) Moderately radiogenic in terms of Sr isotopes: $^{87}Sr/^{86}Sr$ 0.7037–0.7047
(iii) Very unradiogenic in terms of $^{143}Nd/^{144}Nd$: 0.5126–0.5127

While Sr and Pb isotopes of group Ib thus overlap those of the group Ia, there is a distinction evident in terms of Nd isotopes where group Ib lavas appear to have a much less radiogenic signature in comparison to the group Ia lavas.

5.2. Group II — the OIB family of magmas

These magmas are the antithesis of group I magmas in that they are widely spread both spatially and temporally, and are of lower volume. These magmas are all characterized by a relative enrichment in the most incompatible trace elements with a prominent negative K anomaly. These magmas have steeply sloping patterns where the more incompatible elements are ~50–100 more enriched than primitive mantle, while the more compatible trace elements are ~5 times more enriched than primitive mantle (Fig. 20). These magmas universally exhibit radiogenic values of $^{206}Pb/^{204}Pb$ (18.7–20.3) that approach the HIMU mantle reservoir in some cases. However, there is no single isotopic endmember defined by group II lavas.

There are some spatial variations evident in this family of magmas which transgress the subdivision below. Group II magmas in Turkana exhibit less radiogenic values of $^{87}Sr/^{86}Sr$ (0.7031–07034) in comparison to more northerly equivalents (0.7036–0.7044). While all group II magmas are enriched in TiO_2, northerly group II magmas have

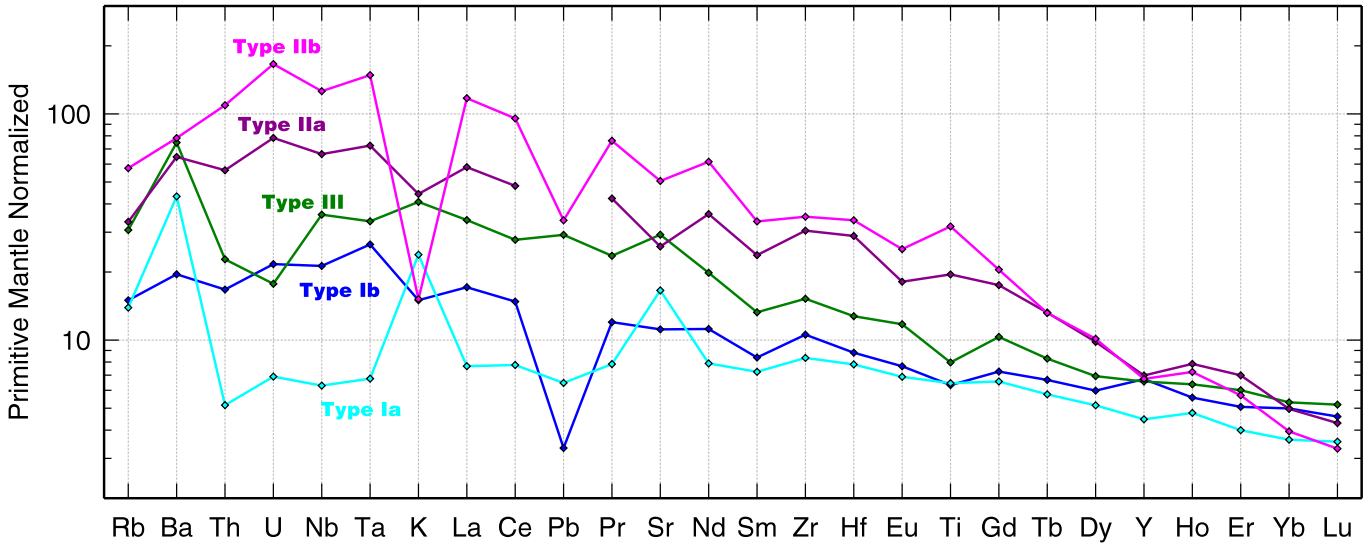


Fig. 20. Primitive mantle normalized magmas (McDonough and Sun, 1995) of lavas that typify the different magma 'families' identified in East Africa. All samples are between 7 and 8 wt.% MgO to facilitate comparison. Data sources: Type Ia (sample 107: Kieffer et al., 2004); Type Ib (sample RG93-73: George and Rogers, 2002); Type IIa (sample LAL-3: Natali et al., 2016); Type IIb (sample 3235: Rooney et al., 2017); Type III (sample 259: Kieffer et al., 2004).

significantly higher values (up to 6.5 wt.%) in comparison to those from Turkana (up to 4 wt.%). A number of sub-members to this family of magmas have been identified on the basis of major and trace element variability:

5.2.1. Group IIa — the MgO-rich compositions

The enrichment in the most incompatible trace elements varies between localities for group II magmas. In particular, early Oligocene HT2 lavas from the northwest Ethiopian Plateau, and Early Miocene magmas from Jarigole (Kenya) exhibit only a 50× enrichment over primitive mantle of the most incompatible trace elements. These localities are also unusual in having picrites, a number of which are the result of significant olivine accumulation (Furman et al., 2006b; Natali et al., 2016; Pik et al., 1999). Group IIa lavas (excluding those that have accumulated olivine) typically have moderate SiO_2 values between 44 and 50 wt.% and CaO of 9–11 wt.%. These lavas are interpreted to have been derived from a warm mantle with a T_P of 1513 °C (Rooney et al., 2012b).

There is profound isotopic variation within this group, with lavas from Turkana consistently having more radiogenic $^{206}Pb/^{204}Pb$ and plotting below the line representing the ε_{Hf}–ε_{Nd} mantle array approaching the HIMU mantle reservoir (Fig. 21). Furman et al. (2006b) suggest that this magma type is derived from melting of a heterogeneous upwelling plume, whereby the HIMU-like magmas are generated from melting of the lithologies convected into the melting zone by the upwelling plume. Natali et al. (2016) suggest the HT2 lavas from the northwest Ethiopian Plateau were generated by melting of Pan-African eclogite components in the upwelling Afar plume. However, in contrast to the HIMU-like signature evident in Kenya, magmas erupted on the northwest Ethiopian Plateau exhibit isotopic characteristics more similar to the "C" mantle reservoir, including moderately radiogenic $^{206}Pb/^{204}Pb$ values, mantle array-like ε_{Hf}–ε_{Nd} (Fig. 21), and elevated $^{3}He/^{4}He$ values of up to 19 R/R_A (Natali et al., 2016; Pik et al., 2006). An alternative model was proposed by Beccaluva et al. (2009) whereby melts from the upwelling mantle plume metasomatized the lithospheric mantle (e.g., Jung et al., 2005), and were melted shortly thereafter to erupt as HT2 flood basalts. In this model, the isotopic characteristics of the HT2 lavas are derived from the "C" plume component, while the major and trace element systematics are imparted by the modal mineralogy of the metasome.

These plume-related modes of origin may not be unique as this magma type is also observed distal from the potential plume source. In middle Egypt, contemporaneous with the rifting of the Red Sea at ca. 24 Ma, group IIa magmas are evident, perhaps suggesting a more complex origin for this signature that is more widespread than is possible through plume–lithosphere interaction (Abu El-Rus and Rooney, 2017).

5.2.2. Group IIb — the carbonated compositions

A distinctive silica-undersaturated group of magmas (35–45 wt.% SiO_2) with a concomitant enrichment in CaO (up to 16.5 wt.%) have been identified during the Eocene in the Turkana basin (Balesa Koromto Type B: Furman et al., 2006b; Rooney, 2010), during the Early Oligocene

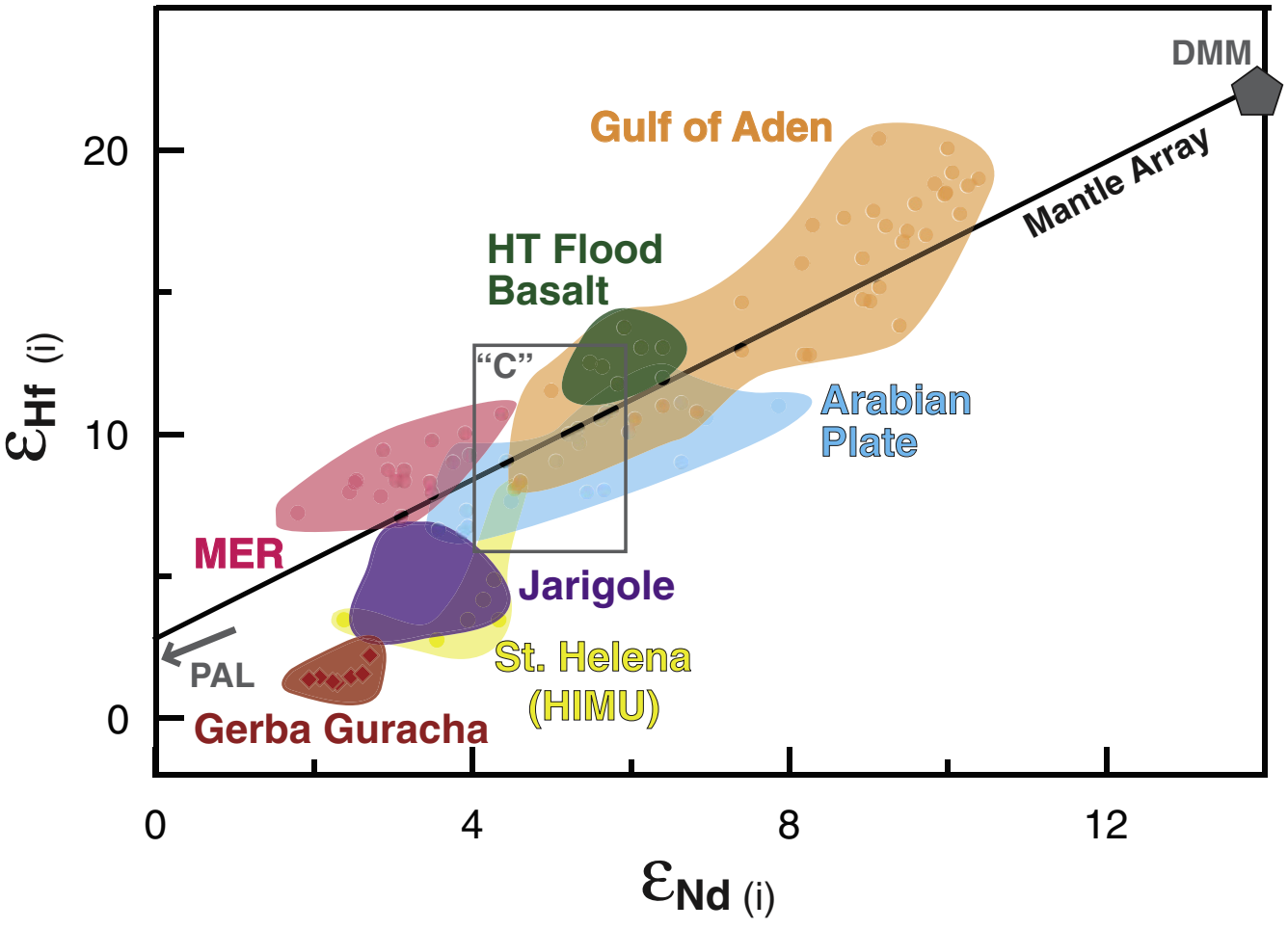


Fig. 21. Age-corrected isotopic variation of regional lavas in terms of Nd and Hf isotopes. Mantle Array is that of Vervoort et al. (1999); PAL — Pan-African lithosphere end member; DMM — Depleted Mantle end member (Rooney et al., 2012a); "C" — C mantle component (Nelson et al., 2012; Rooney et al., 2014b).
Data sources: Gerba Guracha (Rooney et al., 2014b): MER — Main Ethiopian Rift and Gulf of Aden (Furman et al., 2006a; Rooney et al., 2012a; Schilling et al., 1992): Jarigole and HT Flood Basalts (Nelson et al., 2012): Arabian Plate (Bertrand et al., 2003): St. Helena — see Rooney et al. (2014b) for data compilation.

in Yemen (HT2: Natali et al., 2016), and during the Early Miocene on the northwest Ethiopian Plateau (Gerba Guracha: Rooney et al., 2014b, 2017). These suites tend to be more enriched in the most incompatible trace elements in comparison to group IIa lavas. The characteristics of these lavas suggest a role for carbonate in the source of these lavas. It is unclear how a model of a recycled eclogite, convected towards the surface by an upwelling plume (Natali et al., 2016), could be reconciled with the presence of carbonate in the source of such magmas given experimental evidence on carbonated recycled oceanic slabs which suggests the generation of carbon-rich partial melts upon reaching the transition zone that will react with the surrounding peridotite to form diamond (Thomson et al., 2016). While carbonatite melts are frequently associated with plume events, the process is typically considered one where small volume melts separate at depth from an upwelling plume (e.g., Janney et al., 2002). It is unclear why the proposed eclogite component could be consistently associated with such carbonatite melts.

An alternate model is one where these silica under-saturated melts are derived from a metasome within the lithospheric mantle. In this instance, the percolation of melt through the lithospheric mantle can result in a lithology that is rich in amphibole, Fe–Ti oxides and perhaps carbonate (Pilet et al., 2010). Experimental studies on such compositions show that the melting of such a metasome would produce a magma similar to group IIb (e.g., Pilet et al., 2008). Importantly, the results of these experiments show that a garnet-free lithology (i.e. hornblendite) can yield a significantly fractionated REE pattern without the need for garnet in that lithology (Pilet et al., 2008). This is an important observation as it nullifies arguments that deep melting is required in the direct petrogenesis of such magmas on the basis of a REE 'garnet-signature' (Rooney et al., 2017). The low concentration of Ni at Gerba Guracha, in particular, is supportive of a model whereby these group IIb magmas are derived from an olivine poor (or possibly absent) source within the lithospheric mantle (Rooney et al., 2017).

The isotopic characteristics of the Gerba Guracha magmas are quite distinctive in comparison to that of the HT2 flood basalts. Specifically, the Gerba Guracha lavas trend more towards HIMU and not the "C" mantle reservoir as demonstrated by their location on the ε_{Hf}–ε_{Nd} array (Fig. 21). These observations preclude a common model for the origin of these magma types. Instead, variations in Nd, Hf, and Pb isotopes are consistent with in situ evolution from a source identical to those identified during the Pan-African subduction event that constructed the lithosphere in this region (Rooney et al., 2014b). Consequently, a plausible alternative model for the formation of group IIb magmas is through thermo-baric destabilization of a lithospheric metasome that was emplaced during the original construction of the lithosphere in this region (Rooney et al., 2014b, 2017).

Given these observations, it is probable that group II lavas formed through multiple magma generation processes. Common to these is the influence of eclogite/hornblendite in their sources which results in the common major and trace element signature. In contrast, the heterogeneous isotopic compositions likely relate to the magmas which formed the parent eclogite/hornblendite lithologies and the length of time these lithologies have been isolated in the lithosphere or asthenosphere. Thus, while the Oligocene HT2 lavas from the northwest Ethiopian Plateau appear plume-derived, those from Gerba Guracha appear to be lithosphere derived. It is probable that a broad spectrum exists between these two endmembers and that group II lavas might be derived through either process. This permissive definition of the origin for these magmas facilitates an explanation for such magmas occurring beyond the immediate influence of the Afar plume (e.g., Abu El-Rus and Rooney, 2017).

5.3. Group III — the moderate family of magmas

The moderate family of magmas is typified by the more primitive members of the Oligocene HT1 group and the subsequent shield volcanoes of Choke and Guguftu (Fig. 20). In comparison to groups I and II, group III lavas are less immediately obvious in that they do not display the enrichment of group II nor the depletion of group I. This family of magmas is characterized, in primitive mantle normalized diagrams, by a positive Ba anomaly, a negative Th–U anomaly and a positive Nb–Ta anomaly. This characteristic pattern is present in most lavas from this family but can be obscured by assimilation and magma mixing. Magmas from this group have a moderately steep trace element pattern, caused by a significant enrichment in the more incompatible trace elements in comparison to group I. This lava type is widespread and is found throughout the Cenozoic history of East Africa (Rooney et al., 2014a). These lavas are considered melts of a hybrid upper mantle containing components of the depleted upper mantle, the Afar Plume, and continental lithospheric contamination (e.g., Rooney et al., 2013). Isotopically, the moderate family of magmas are very diverse and do not have a characteristic range. The isotopic composition of a lava from this group is controlled by the ratio of the upper mantle, plume, and lithospheric endmembers which comprise the melt (Fig. 10). Thus, the lavas can occur within a ternary mixing space defined by these endmembers.

The broad family groupings I have identified here should not be viewed as exhaustive, but rather a top level division of magmas from which others may be identified. Most lavas described here fall into one of the three groupings but ambiguity remains as transitional lava suites exist (e.g. Gamo). In reality, some degree of mixing between sources and melting mechanisms may result in an array of magmas with various geochemical signatures — I contend that the endmembers of these hybrid magmas would fall into one of the groups outlined herein.

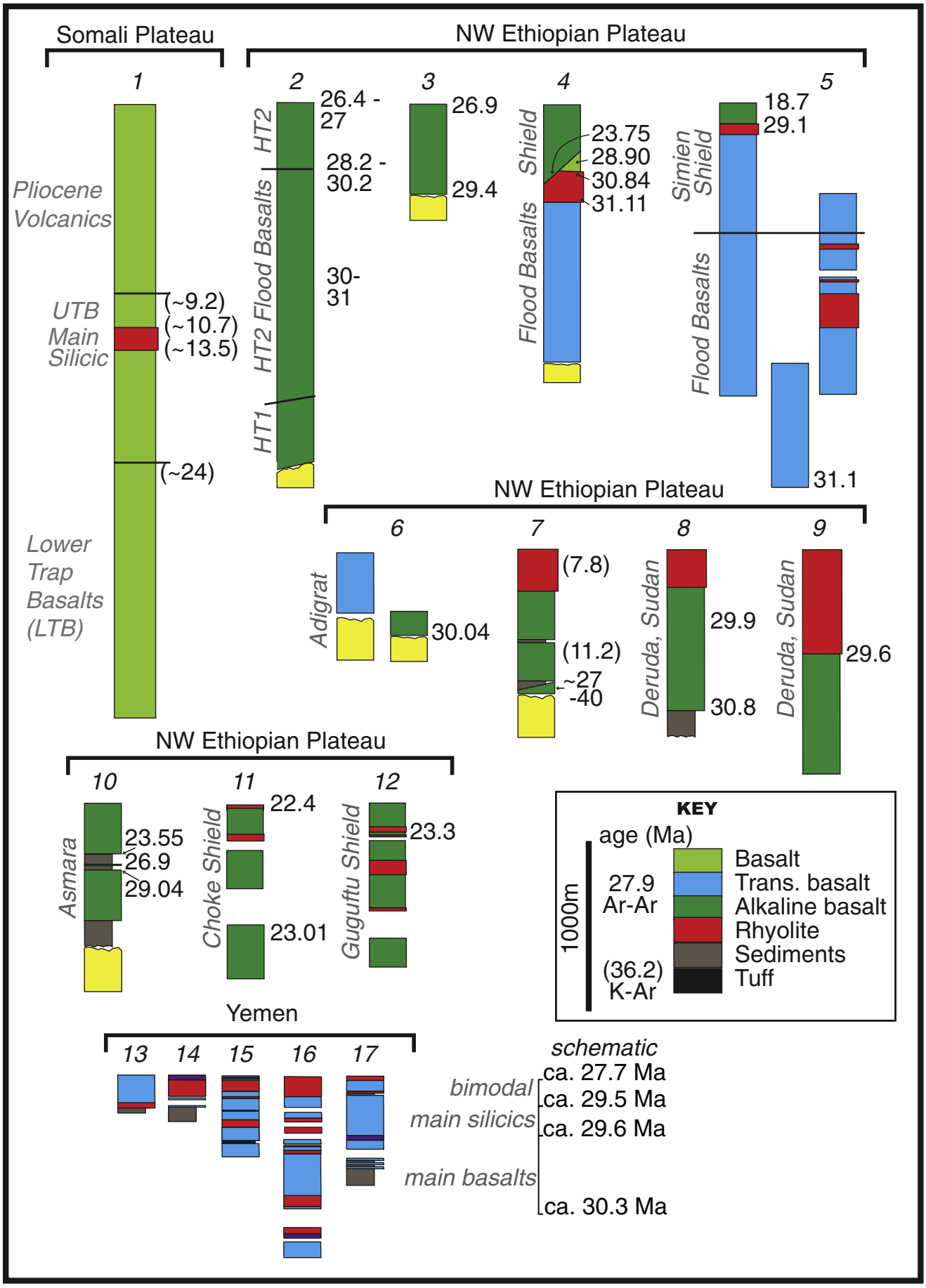
6. Thermo-chemical anomalies in the East African mantle

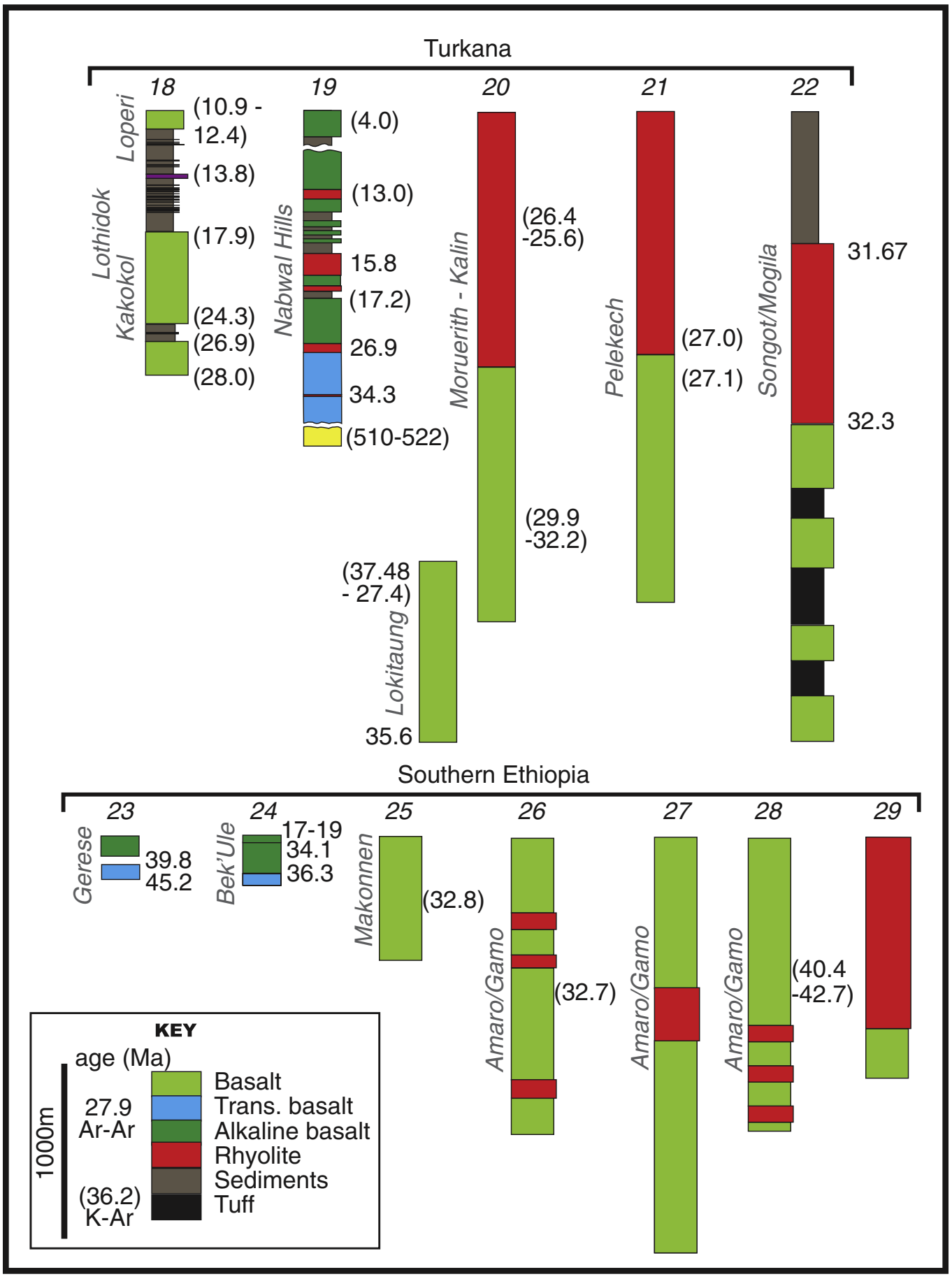
6.1. Evidence of an anomalous upper mantle during the Cenozoic magmatic activity

From the Eocene to the Miocene, thermo-chemical anomalies in the upper mantle have been central to magma generation in East Africa. Despite the consensus among the geochemical community that this material was derived from anomalous domains at some depth within the mantle, the precise source, composition, and number of such anomalies remains hotly debated. The modern conditions of the East African upper mantle, as revealed by seismic tomography, suggest that it contains a significant volume of melt (Bastow et al., 2005, 2008, 2010), which may be segmented beneath Afar due to the effects of decompression associated with the development of a proto-spreading center (Gallacher et al., 2016). While some geophysical studies suggest the mantle beneath modern Afar is not influenced by a thermo-chemical anomaly (Hammond et al., 2013; Rychert et al., 2012), geochemical and numerical modeling constraints clearly show the influence of elevated mantle potential temperature in this region (Armitage et al., 2015; Ferguson et al., 2013; Rooney et al., 2012b).

The upper mantle beneath East Africa is described more broadly as having anomalously slow P-wave velocity over a wide area (400 km deep by 500 km wide: Benoit et al., 2003, 2006; Bastow et al., 2008), that cannot be entirely thermal in origin (Rooney et al., 2012b), and which may connect to a lower mantle anomaly (Hansen et al., 2012).

Fig. 22. Schematic stratigraphic columns for the regions indicated by numbers on the map represented in Fig. 3. The length of each column represents the thickness of material erupted as noted by the scale bar. ^{40}Ar–^{39}Ar ages are noted by numbers beside the columns, while numbers in brackets are K–Ar ages. The age reported is placed at the approximate position within the column where it was measured. For Yemen, a schematic column of ages is reported that is not linked to the stratigraphy shown. Data sources: (1) Juch (1978). This column is the average thicknesses given for the units presented in that work. More detailed columns along the margin are presented in that work that are not shown here; (2–3) Pik et al. (1998, 1999) and dates from Hofmann et al. (1997). (4) Prave et al. (2016); (5) Kieffer et al. (2004); (6) Beccaluva et al. (2009); (7) Berhe et al. (1987); (8–9) Kenea et al. (2001); (11–12) Kieffer et al. (2004); (13–17) Chiesa et al. (1989); Ukstins-Peate et al. (2005).


Somali Plateau
NW Ethiopian Plateau
1
2
3
4
5
Pliocene Volcanics
UTB
Main Silicic
Lower Trap Basalts (LTB)
(~9.2)
(~10.7)
(~13.5)
(~24)
HT2
HT2 Flood Basalts
HT1
26.4 - 27
28.2 - 30.2
30-31
26.9
29.4
Shield
Flood Basalts
23.75
28.90
30.84
31.11
Simien Shield
Flood Basalts
18.7
29.1
31.1
NW Ethiopian Plateau
6
7
8
9
Adigrat
30.04
(7.8)
(11.2)
~27
-40
Deruda, Sudan
29.9
30.8
Deruda, Sudan
29.6
NW Ethiopian Plateau
10
11
12
Asmara
23.55
26.9
29.04
Choke Shield
22.4
23.01
Guguftu Shield
23.3
KEY
age (Ma)
1000m
27.9 Ar-Ar
(36.2) K-Ar
Basalt
Trans. basalt
Alkaline basalt
Rhyolite
Sediments
Tuff
Yemen
13
14
15
16
17
schematic
bimodal
main silicics
main basalts
ca. 27.7 Ma
ca. 29.5 Ma
ca. 29.6 Ma
ca. 30.3 Ma


Turkana
18
19
20
21
22
Loperi
Lothidok
Kakokol
(10.9 - 12.4)
(13.8)
(17.9)
(24.3)
(26.9)
(28.0)
Nabwal Hills
(4.0)
(13.0)
15.8
(17.2)
26.9
34.3
(510-522)
Moruerith - Kalin
(26.4 -25.6)
(29.9 -32.2)
Pelekech
(27.0)
(27.1)
Songot/Mogila
31.67
32.3
Lokitaung
(37.48 - 27.4)
35.6
Southern Ethiopia
23
24
25
26
27
28
29
Gerese
39.8
45.2
Bek'Ule
17-19
34.1
36.3
Makonnen
(32.8)
Amaro/Gamo
(32.7)
Amaro/Gamo
Amaro/Gamo
(40.4 -42.7)
KEY
age (Ma)
1000m
27.9 Ar-Ar
(36.2) K-Ar
Basalt
Trans. basalt
Alkaline basalt
Rhyolite
Sediments
Tuff

Evidence of a lower-mantle anomaly beneath southern Africa which extends from the core–mantle boundary to ~1100 km depth might connect to the anomalies imaged under East Africa (Hansen et al., 2012; Mulibo and Nyblade, 2013). These assertions are consistent with S- and P-wave travel time anomalies beneath Afar and the Main Ethiopian Rift, suggesting a thermal and chemical perturbation of the mantle transition zone (Civiero et al., 2015; Cornwell et al., 2011; Julià and Nyblade, 2013; Thompson et al., 2015). While these seismic investigations of the state of the modern mantle cannot constrain the thermo-chemical conditions during the Eocene-Miocene, elevated mantle potential temperature (Rooney et al., 2012b), and isotopic anomalies that could not be derived from the upper mantle or lithosphere (e.g., Pik et al., 2006), are a consistent feature of the East African upper mantle throughout the Cenozoic. It is thus apparent that the LLSVP at the core–mantle boundary beneath a static Africa (Garnero and McNamara, 2008), which is a persistent mantle feature over timescales much larger than that in question (e.g., Torsvik et al., 2008), could have contributed to magmatism in East Africa during the Cenozoic.

6.2. Geometry and structure of the mantle thermo-chemical anomaly

The concept of a mantle plume, which I describe here as a thermo-chemical anomaly derived from a thermal boundary layer without prejudice as to its geometry, is firmly established within the geochemical literature in East Africa. The derivation of material from sources deep within the mantle is consistent with the geophysical and geodynamic studies outlined in the previous paragraph, but the manner in which this material ascended through the lithosphere remains a topic of active debate. Existing studies have focused on three broad models: (A) a single thermo-chemical anomaly responsible for magmatism throughout this period, (B) two distinct thermo-chemical anomalies — one in the south and one in the north, (C) two branches of the same thermo-chemical anomaly which influence magmatism in the north and south of the region.

6.2.1. Single thermo-chemical anomaly

The model of a single thermo-chemical anomaly was described by Ebinger and Sleep (1998), whereby a single large plume head impacted beneath the Ethiopian Plateau and channeled material along the base of the lithosphere to other regions. Ebinger and Sleep (1998) suggest that variations in the thickness of the continental lithosphere resulting from prior rifting events may have facilitated the flow of material from this single plume. Ponded plume material would then melt to create the range of magmas observed. Increases in magma flux to the surface would have coincided with episodes of lithospheric extension and consequent decompression melting.

6.2.2. Two thermo-chemical anomalies

In contrast, a two plume model was put forward in order to explain the geochemical heterogeneities between rocks in the southern portion of the region and that in the north (George and Rogers, 2002; George et al., 1998). This model resulted in the concept of a 'Kenya plume' and 'Afar plume'. On the basis of potential plate reconstruction models, it was suggested that the Eocene activity in Southern Ethiopia was most likely the result of activity associated with the Kenya plume, while Oligocene activity on the northeastern Ethiopian Plateau was attributed to the Afar plume (George and Rogers, 2002).

6.2.3. Multiple branches from a common source

This two plume concept was modified by Furman et al. (2006b) who suggested that the Kenya plume and Afar plume were both derived from the same common source — the South African Superplume (now broadly discussed as the African LLSVP). Variations in the geochemical characteristics of lavas from the two regions were attributed to recycled materials carried within the ascending plumes and to the inherent heterogeneity of the African Superplume itself (Furman et al., 2006b). This model was modified by Meshesha and Shinjo (2008) on the basis of their identification of four components within the plume and resolved this observation by suggesting plumelets within a broader upwelling. However, the origin of these components (i.e. within a broad upwelling zone or from the continental lithosphere) remains contentious.

6.3. Insights into the composition of the African LLSVP

The developing consensus that material upwelling beneath East Africa is derived from a common deep source has been strengthened by observations of the isotopic characteristics of noble gases in the region which are consistent with a single origin within the African Superplume (Halldórsson et al., 2014). Variability in the other isotopic systems may result from different components sampled from the LLSVP. Os-Hf isotopic data pointed to some potential distinctions between the plumes (Nelson et al., 2012; Rogers et al., 2010). Magmas thought derived from the Afar plume lack Os isotopic evidence for the recycling of a subducted slab (Rogers et al., 2010), and are likely derived from ambient mantle in the vicinity of the LLSVP (Nelson et al., 2012).

Despite these insights, the influence of the African lithospheric mantle has had a confounding impact on the efforts to constrain the composition of the upwelling plume and its deep mantle source. Magmas used to probe the plume composition have dominantly been of the group II variety described earlier due to their typically higher Mg#. Given the evidence that such magmas can be derived from the continental lithospheric mantle (Beccaluva et al., 2009; Rooney et al., 2014b, 2017), caution is advised in using such magmas as a probe of the composition of the material upwelling from the LLSVP. Recent models suggest that 'drips' of lithospheric material may (in part) be responsible for the generation of such magmas (Furman et al., 2016), though other models suggest a contribution from eclogite within the upwelling plume (Natali et al., 2016). An important outcome of these studies is that not all components identified within primitive lavas from East Africa need be derived from materials upwelling within the LLSVP. Further studies of groups I and III lavas are necessary in order constrain plume composition, but these lavas tend to be more fractionated and consequently experience interaction with the continental crust (e.g., Baker et al., 1996b; Kieffer et al., 2004).

7. Volume and aerial extent of magmatic activity

The extent and volume of lavas within the Ethiopian Large Igneous province has not been revisited since initial estimates in the 1960s and early 1970s (Baker et al., 1972; Mohr, 1963). These initial studies suggested that the Ethiopian flood basalt province (excluding Yemen) occupies a current area of 600,000 km^2, but was thought to have initially covered an area of some 750,000 km^2 (Mohr, 1963). Baker et al. (1972) suggest a total magmatic volume of 350,000 km^3, however, the wide

Fig. 23. Schematic stratigraphic columns for the regions indicated by numbers on the map represented in Fig. 3. The length of each column represents the thickness of material erupted as noted by the scale bar. ^{40}Ar–^{39}Ar ages are noted by numbers beside the columns, while numbers in brackets are K–Ar ages. The age reported is placed at the approximate position within the column where it was measured.
Data sources: (18–21) Brown and McDougall (2011); McDougall and Brown (2009); McDougall and Watkins (2006); Tiercelin et al. (2012); (22) Wescott et al. (1999); Brown and Jicha (2016); (23–24) George et al. (1998); George and Rogers (2002); (25–29) Davidson (1983). The sections in this area are divided into the Makonnen basalts and the older Main Series volcanic rocks that have been renamed herein as Amaro/Gamo.

range in thicknesses observed for the Oligocene sequences, deep erosion around the plateau margin, and the increased areas noted above made this a minimum estimate. Over the past 50 years, geologic and chronologic investigations of East Africa have resulted in revised maps and a more comprehensive understanding of both the distribution of magmatic activity and the linkage between the Yemen and Ethiopian Traps to form a single African–Arabian Large Igneous Province. A revision of the lava aerial distribution and volume estimates is therefore necessary.

Using prior estimates of volume and area of the Ethiopian LIP, the region lies off of the global array of volume to area continua for most other Phanerozoic flood basalts (Stein et al., 2015, 2016), suggesting that either an unusually low volume of basalt was erupted or that this material was more widely distributed. This may also have in part been due to what the prior estimates included (rifted areas) and excluded (Yemeni Traps). I have undertaken a revision to this initial work on the basis of our new knowledge of the stratigraphy of the province (Figs. 22, 23) (see Supplemental information for details). My estimated total aerial extent of lavas (pre-erosional) associated with the African–Arabian LIP is 720,000 km^2, with an estimated volume of also ~720,000 km^3 (Table 1). These data, suggest an average lava thickness of ~1 km, more consistent with other Phanerozoic LIPs (Stein et al., 2015), though these volume estimates do not significantly alter the relative magnitude of the province, which lies between the Emeishan and Deccan in terms of volume and area.

The volume and area estimates presented herein have some important provisos: (A) I have amalgamated all magmatic activity in the region into a single LIP. There is some discussion on the relative role of thermo-chemical anomalies associated with Kenyan and Afar plumes during this time (e.g., George and Rogers, 2002), but I have not attempted to subdivide the magmatic activity due to the likely common source of these thermochemical anomalies (Furman et al., 2006b). (B) Not included within these estimates, is magmatic activity within the East African Rift system and Afar. While the Afar, and potentially the Kenya plumes continue to influence magmatism in extending areas throughout the region (e.g., Rooney et al., 2012a), the primary cause of this magmatism relates to rifting and I have thus not included this magmatism with the African–Arabian Large Igneous Province.

8. Conclusion

From ca. 45 to 20 Ma the 720,000 km^3 of magmatism in East Africa was dominated by three distinct pulses of basaltic magmatism, each followed by bimodal or silicic magmatic activity:

(A) Eocene Initial Phase: The initial pulse of Cenozoic magmatism in East Africa is centered on the Omo region of Southern Ethiopia. Here, large volumes of magma were erupted over a ~10 Ma window from 45 to 35 Ma, though the precise timing of the event remains poorly constrained. Further afield, less volumetrically significant magmatism in Turkana appears synchronous with this initial event. The origin of this magmatic event is considered related to mantle plume activity, but possible lithospheric contamination of the lavas from this period have thus far prevented robust comparison with the subsequent Oligocene Traps event.

(B) Oligocene Traps Phase: The second pulse of magmatism in East Africa is centered on the NW Ethiopian Plateau and the formerly contiguous portions of Yemen. Here, the timing of the initiation of this event at ca. 31.2 ($\pm$0.6, 1σ) Ma (Hofmann et al., 1997) is better constrained than the initial event in Southern Ethiopia, but establishing the precise date of commencement is elusive (Ukstins et al., 2002). A new precise zircon age of 31.108 ($\pm$0.041, 1σ) Ma recently published for rhyolites overlying the main flood basalt series (Prave et al., 2016), suggests the initiation of basaltic magmatism might have preceded the currently accepted 31.2 Ma date. After the initial mafic pulse, magmatism became bimodal, and eventually silicic in both Ethiopia and Yemen. The origin of this event is considered to result from the interaction of the Afar mantle plume activity with the African lithosphere.

(C) Early Miocene Resurgence Phase: The final magmatic pulse that occurs during this period appears to be synchronous over a wide area, potentially from the Red Sea margin to Turkana. Basaltic magmatism is evident as flows over paleosols and volcanoclastic sediments of the preceding magmatic cycle. These magmas typically date close to 23–24 Ma and are well constrained due to the importance of these flows in constraining fossil horizons (Abbate et al., 2014; Boschetto et al., 1992). This final pulse appears to be volumetrically less significant in comparison to the two preceding it, in that lavas erupted during this period did not build thick basaltic piles and instead dominantly manifested as shield volcanoes or isolated flows. The origin of magmatism during this event remains unclear, with evidence of lithospheric melts and contributions from the Afar plume, coupled with synchronous extension of the continental crust (Stab et al., 2016).

Given the almost stationary (O'Connor et al., 1999) or very slowly migrating African plate (Ebinger and Sleep, 1998), it is tempting to suggest that the pulsed nature of the Cenozoic magmatic activity in East Africa relates to parallel pulses of material rising from the African LLSVP. However, such a linkage is not assured, as the thick African continental lithosphere provides a mediating influence on melt production within the African upper mantle. Specifically, given the evidence of elevated mantle potential temperatures (Rooney et al., 2012b), and the unusual isotopic signatures found in magmas throughout the Cenozoic (Furman et al., 2006b; George and Rogers, 2002; Kieffer et al., 2004; Rooney et al., 2013), a viable alternative model is where the African upper mantle is continuously influenced by the African LLSVP. In such circumstances, magmatic pulses could relate to extensional events within the continental lithosphere that facilitate decompression melting of the plume-influenced upper mantle. In reality, the solution to the observations presented herein could lie between these two endmembers. The synchronous region-wide magmatic events described in this contribution from 45 to 22 Ma are perhaps suggestive that variable material flux from the African LLSVP is the more dominant control on magma generation. Subsequent to 22 Ma, magmatic events become localized and are clearly associated with lithospheric extension, becoming more focused on the developing rift. It is the potentially dynamic relationship between magma generation, the flux of material from the African LLSVP, and lithospheric extension, that serves as a convenient breaking point for this first part of my review of Cenozoic East African magmatism.

Table 1
Area and volume estimates for African–Arabian Large Igneous Province. Area estimates are derived from polygons drawn in Google Earth. Discussions on Lava thickness estimates and geometry are discussed in the online supplement text.

Sub-region	Area of polygon (km^2)	Avr. lava thickness (m)	Total volume (km^3)
NE Ethiopian Plateau	274,395	1600	439,032
SE Ethiopian Plateau & Chew Bahir Basin	299,324	500	149,662
Turkana Basin	22,317	1000	22,317
Somali/Harrar Plateau	48,389	650	31,453
Yemen	77,108	1000	77,108
Total for the Province	721,533	–	719,572

Acknowledgements

Rooney is funded by the United States National Science Foundation through grants EAR 1551872 and EAR-1219647. I would like to thank Andrew Kerr for his tireless editorial handling of this contribution. Detailed and constructive reviews by Stefan Jung and an anonymous reviewer helped improve the manuscript. I am grateful to Bernard Le Gall for providing the revised geological map of Djibouti. Discussions with Daniel Mège and Peter Purcell helped frame my understanding of magmatism in Ogaden. Wendy Nelson is thanked for providing details of magmatism in the Bale Mountains.

Appendix A. Supplementary data

Supplementary data to this article can be found online at http://dx.doi.org/10.1016/j.lithos.2017.05.014.

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